



Orion: Beyond the Frontier

Version: 1

Table of contents

- What is Orion: Beyond the Frontier** 4
 - Game Principles 5
 - A word about “realism” 6
 - Safety Tools 6
- Characters** 9
 - Details 10
 - Playing Synthetic and Uploaded Characters 15
- Gear** 17
 - Tools 17
 - Weapons 17
 - Armor 18
 - Vehicles 18
 - Cybernetics 19
 - Miscellaneous 19
- How to play** 21
 - Rolling 22
 - Load 22
 - Advancement 23
 - Damage Control 23
 - Injuries 24
 - Game mastering 24
 - Playing Solo 26
- Interstellar Travel** 31
 - Navigation Hazards 31
 - Core Technologies for Travel 32
 - Skills for Interstellar Success 33
 - Travel Time and Fuel Consumption 34

Interplanetary Travel	34
Environmental Hazards	36
Vehicles of the Orion Frontier	39
Starships	43
Starship Types and Classes	44
Combat	48
Shipyard	49
The Orion Frontier	55
The Age of Orion (2400–4000)	56
Culture and Society	60
Psionics and the Resonant Tradition	64
Major Stellar Polities	65
Sol Union	65
Orion Concordium	66
Sirius Compact	66
Nova Terra Assembly	67
Zal-Terran Commonwealth	68
Frontier Coalition	69
Factions and Alliances	71
Major Factions	71
Minor Factions	76
Cosmic Threats	81
Faction Standing	83
Galactic Commons	85
Types of Galactic Commons	85
Key Features of the Galactic Commons	87
Using the Galactic Commons in Gameplay	87
Famous Locations in the Galactic Commons	88
Living the Frontier	91
Daily Life on the Frontier	91
Challenges of Frontier Life	92
Aesthetics of Frontier Towns	94
Survival Tactics	94
Using the Frontier in Gameplay	95
Precursor Echoes	97
Cultural Significance of the Precursors	97

The Ark of Precursor Knowledge	98
Exploring Precursor Ruins	98
Precursor Artifacts	99
Mysteries of the Precursors	100
Random Artifact Table	100
Stellar Fauna and Flora	103
Flora	103
Fauna	104
Ecosystems in the Frontier	105
Using Flora and Fauna in Gameplay	106
Macro-Scale Hazards	107

The atmosphere within the *Radiant Horizon* was tense. As the ship cruised along the dim expanse of interstellar void, its hull bathed in the faint glow of a distant pulsar, the team assembled on the observation deck. Beyond the reinforced viewport, the nebulous silhouette of a derelict Precursor megastucture loomed—an ancient and colossal artifact orbiting a decaying star.

“Scans are picking up anomalies all over that thing,” said Dr. Ilyan Jassir, the ship’s Archivist. His voice quivered slightly, not from fear, but from awe. “This could predate the Ark of Precursor Knowledge by a million years. The energy signatures... They’re impossible to categorize.”

Captain Maia Veyron leaned against the console, her eyes narrowing. “And the radiation spikes? Are we safe to proceed?”

Ilyan hesitated. “With shielding at maximum and... no sudden movements, yes. But the structure itself is unstable—gravitational shifts could trap us if we’re not careful.”

At the center of the deck, Guardian Rook tightened his grip on his pulse rifle. “Let me guess. We’re going in anyway.” His deep voice carried a note of reluctant acceptance.

“We didn’t travel this far for a stellar light show,” Maia replied curtly. She turned to Tamsin Corwin, the Tech-Scavenger. “Corwin, are you ready to breach the outer shell?”

Tamsin grinned, flipping her visor down. The intricate wiring of her exosuit glinted as it synced with her neural implants. “Always. If those ancient bastards left anything valuable behind, I’ll find it. You just keep the anomalies from frying my brain.”

As the crew suited up, the *Radiant Horizon* shuddered slightly—a reminder of the unstable space around the artifact. Maia tapped her comm unit. “Horizon Pilot, keep the engines on standby. If this goes south, I want to be out of here in under five seconds.”

“Copy that, Captain,” replied Ezra Kael from the helm. “But fair warning: Subspace anomalies are making FTL calculations a nightmare. If we need to run, it’ll be rough.”

“Rough is better than dead,” Maia replied before cutting the comm link.

The docking tube extended slowly, connecting the *Radiant Horizon* to the yawning maw of the Precursor megastructure. Through her visor, Maia watched as Tamsin moved first, her magnetic boots clanking against the structure’s ancient surface. Dr. Jassir followed, his portable scanner flickering with strange data streams.

“This place...” Jassir whispered, his voice crackling over the comms. “It’s not just a ruin. It’s a graveyard.”

Rook was the last to step through, his gaze scanning every shadow for movement. “Then let’s not add our names to the list.”

Inside, the crew found themselves in a vast, cathedral-like chamber. Walls of crystalline lattice hummed faintly, refracting the light from their helmets into a spectrum of alien hues. Glyphs pulsed softly, as if responding to their presence.

“These are neural relays,” Jassir said, kneeling beside a glowing console. “The Precursors communicated directly with this structure. It’s... alive, in a way.”

“Alive?” Tamsin muttered, her tone skeptical. “Great. Let’s poke it and see if it bites.”

Before anyone could respond, the chamber trembled. A low, resonant hum filled the air, and a section of the lattice shifted, forming what resembled a doorway.

“Captain...” Rook’s voice was tight. “We’ve got movement.”

Maia tightened her grip on her sidearm. “Stay sharp. Corwin, Jassir—analyze whatever you can, but be quick. We’re not staying longer than necessary.”

The doorway flickered, and through it came a wave of heatless light. Within the radiance, shapes began to form—abstract at first, but gradually coalescing into something disturbingly humanoid.

“It’s a projection,” Jassir stammered, stepping back. “Some kind of... interface.”

The figure spoke, its voice reverberating like a choir from a distant star. “You tread on the bones of gods. Speak your purpose, travelers, or be unmade.”

Maia stepped forward, her heart pounding. “We’re here to learn. To understand.”

The projection paused, its form shifting into a kaleidoscope of shapes. “Understanding has a cost.”

Tamsin’s voice cut through the tension. “Yeah, well, let’s hope it’s payable in credits.”

As the projection moved closer, its light began to pierce the crew’s suits, the glyphs on the walls glowing brighter. Maia felt a pressure in her mind—a presence sifting through her thoughts.

“Captain,” Ezra’s voice crackled over the comm. “The anomaly’s destabilizing. We need to extract now!”

“Jassir, Corwin, move!” Maia barked, retreating toward the docking tube. But before they could leave, the projection extended a hand—solidifying into a crystalline shard. It floated toward Jassir, pulsing with unearthly energy.

“It’s... giving us something,” he murmured, awestruck.

“Take it and run,” Maia ordered.

As Jassir secured the shard, the chamber began to collapse, the walls warping inward as if the structure itself sought to consume them. The crew sprinted back to the *Radiant Horizon*, the airlock sealing just as the megastructure imploded into a singularity of light.

“Punch it, Ezra!” Maia yelled as the ship shuddered violently.

The *Radiant Horizon* shot into slipstream, narrowly escaping the singularity’s pull. In the silence that followed, the crew gathered on the deck, their breaths heavy.

Jassir held the shard up to the light, its crystalline surface shifting with an otherworldly glow. “This... this could change everything.”

Maia nodded, her gaze fixed on the distant stars. “Or it could destroy us. Either way, it’s not the end, it’s just the beginning.”

What is Orion: Beyond the Frontier

Orion: Beyond the Frontier is a roleplaying game in which you will define an imagined space bounded by rules. Actions and events occur through procedure-driven player interaction. This facilitates an emergent shared narrative.

Put plainly, players will assume the roles of characters in a fictional setting. The characters will act according to the rules of the game that will guarantee their success or failure. The outcome of actions will have an effect on the narrative that emerges.

A game of *Orion: Beyond the Frontier* takes place as a conversation between the Referee and the other players.

- The Referee sets up the situation, describes the environment and setting, moves the dialogue and actions of the Non-Playing Characters.
- The Players move in the imagined space of the game through their Playing Character, describing their actions and narrating their conversations. It is up to each player to decide whether to act out their Character or play them impersonally. Players should also ask questions of clarification to the Referee.

To play you will need:

- At least 2 players, one will cover the role of Referee. The ideal number of players is 4-5 players, maximum 6.
- A complete set of polyhedral dice, preferably one for each player.
- Writing instruments, index cards and sheets of paper for note-taking.

Game Principles

It is important that everyone at the table feels comfortable and has the following principles in mind:

- **Fiction over mechanics.** This ruleset is intentionally streamlined and prioritizes fiction over complex mechanics. If you are primarily interested in a game with detailed mechanical systems, there are other rulesets that may be better suited to your preferences.
- **The Referee is a player.** Although the Referee has different responsibilities than the other players, they should still be considered as a fellow player and should not shoulder more responsibility for the success of the game than anyone else at the table.
- **Referee's Mediating Role.** The Referee does not compete against other players; their role is solely to arbitrate situations within the context of the game. They are responsible for maintaining consistency, ensuring fairness, and resolving any disputes or conflicts that may arise during play.
- **Promote Cooperative Play.** This game is intended to be played collaboratively, with a focus on teamwork and overcoming challenges as a group. As such, competition between players should be discouraged in favor of cooperation and mutual support. The characters should work together to confront the obstacles and threats posed by the game world, rather than engaging in individualistic or adversarial behavior.
- **Information Query.** Players should ask the Referee questions in a concise and straightforward manner. The Referee, in turn, should provide clear and detailed explanations of the context and the potential risks associated with each action taken by the characters. This helps to ensure that players make informed decisions and understand the consequences of their actions.

- **Action first.** When describing an action, players should first provide a clear and detailed description of what their character is attempting to do. Only after the action has been fully described should the Referee request a dice roll, if necessary, to determine the outcome of the action. This helps to ensure that the narrative flows smoothly and that players have a clear understanding of the sequence of events.
- **Emergent Fiction.** The game thrives on emergent fiction, where the narrative evolves naturally from player choices and interactions. Expect unexpected developments and let the story follow where decisions lead. Deviating from preconceived plots is part of the design.
- **Embrace uncertainty.** Embracing uncertainty is at the core of this game. Embrace the unpredictability of player choices and the twists and turns the narrative may take. Avoid rigidly planning every detail and be open to adapting to the unexpected. Unexpected turns often make for the best sessions.

A word about “realism”

In playing *Orion: Beyond the Frontier*, it is important to embrace a sense of realism that remains consistent with the game’s setting, backgrounds, equipment, and accompanying illustrations. This far-future science fiction universe, grounded in plausible extrapolation but unafraid of space-operatic scale, gives players a specific and consistent setting to work in. Prior knowledge of science fiction is not required, though reading the sources of inspiration adds context. While true “realism” is impossible in a fictional game, avoid indulging in absurdities: strive for consistency with the established setting and with general principles of physics. Keeping to this makes the fiction coherent and the stakes plausible.

Safety Tools

It is important to remember that not everyone at the table may be comfortable with the themes or content addressed by the game, or with the actions of the other players. It is therefore important to establish clear boundaries and expectations at the beginning of the game, and to be sensitive to the needs and feelings of all participants throughout the session. It may also

be helpful to establish a system for communicating discomfort or taking breaks if needed.

There are various safety tools and techniques that can help create a comfortable and safe environment for everyone involved in the game, such as Lines And Veils, C.A.T.S., the X-Card, and Script Change.

It is important to familiarize yourself with these tools and techniques and to use them proactively to ensure that everyone feels safe and comfortable during the game.

Characters

Playable characters are the players' gateway to the game universe in **4000 CE**. By assuming their role, players will shape the narrative through their intentions and actions, navigating the fractured and mysterious realms of the **Orion Frontier**.

1) Choose Your Role

ADVENTURER: Skilled in Piloting (d8) and Survival (d8). Take a modular exploration suit, survival toolkit, and a small atmospheric shuttle.

ARCHIVIST: Skilled in Science (d8) and Computers (d8). Take a portable sensor array, data archives, and a holographic research assistant.

DIPLOMAT: Skilled in Negotiation (d8) and Linguistics (d8). Take cultural exchange documents, a comm suite with encrypted channels, and adaptive attire.

ENGINEER: Skilled in Engineering (d8) and Resource Management (d8). Take a repair drone, multi-tool gauntlet, and a resource processor (bulky).

HORIZON PILOT: Skilled in Navigation (d8) and Piloting (d8). Take a personal spacecraft with enhanced sensors, a flight suit, and a tactical navigation device.

GUARDIAN: Skilled in Tactics (d8) and Firearms (d8). Take a reinforced exo-suit, pulse rifle, and a tactical drone.

MEDIC: Skilled in Medicine (d8) and Xenobiology (d8). Take a medkit, portable diagnostic unit, and genetic repair patches.

MERCHANT: Skilled in Negotiation (d8) and Networking (d8). Take an encrypted market database, portable goods container, and 8 Supply Credits.

SHADOW: Skilled in Stealth (d8) and Psi-Ops (d8). Take a cloaking device, a neural inhibitor, and a concealed weapon of choice.

TECH-SCAVENGER: Skilled in Engineering (d8) and Sensors (d8). Take a magnetic harness, a portable analysis rig, and a quantum artifact container.

2) Pick Three Skill Increases (from no skill->d8->d10->d12:)

Skills Available:

Close Quarters Combat, Computers, Cyber-Ops, Demolitions, Engineering, Exo-Suit Operation, Firearms, Linguistics, Medicine, Navigation, Negotiation, Networking, Piloting, Psi-Ops, Resource Management, Science (specializations: Astrophysics, Nanoengineering, Xenology, etc.), Sensors, Stealth, Survival, Tactics, Xenobiology, Zero-G Maneuvering.

Details

Customize details to fit the setting. Here are some options fitting a hard sci-fi space crew:

Male Names

|||

|—|—|

| 1 | Kael |

| 2 | Orion |

| 3 | Zane |

| 4 | Eryx |

| 5 | Solis |

| 6 | Darien |

| 7 | Alaric |

| 8 | Soren |

| 9 | Jarek |

| 10 | Riven |

| 11 | Lysander |

| 12 | Kaidan |

13	Altair
14	Zephyr
15	Ryker
16	Evander
17	Thane
18	Aedric
19	Malric
20	Caelum

Female Names

—	—
1	Lyra
2	Astra
3	Seris
4	Calla
5	Elara
6	Nyssa
7	Althea
8	Ceryn
9	Veyra
10	Orielle
11	Novara
12	Selene
13	Taryn
14	Kaida
15	Vesper
16	Amara
17	Syra
18	Evyn
19	Naeris
20	Nerys

Neutral Names

|||

|—|—|

| 1 | Rune |

| 2 | Aeris |

| 3 | Cyra |

| 4 | Kiran |

| 5 | Lioren |

| 6 | Quin |

| 7 | Elyon |

| 8 | Zeal |

| 9 | Talis |

| 10 | Korrin |

| 11 | Nyx |

| 12 | Auron |

| 13 | Sio |

| 14 | Evros |

| 15 | Remy |

| 16 | Kaelix |

| 17 | Rhaen |

| 18 | Valin |

| 19 | Daris |

| 20 | Eris |

Surnames

|||

|—|—|

| 1 | Kestrel|

| 2 | Duvall |

| 3 | Kaelor |

| 4 | Orinix |

| 5 | Thorne |

| 6 | Veyron |

| 7 | Talwyn |

8	Nyvar
9	Lannix
10	Ardeen
11	Cyrell
12	Morven
13	Faelix
14	Drayven
15	Velara
16	Zayrus
17	Kyros
18	Aelric
19	Rhyne
20	Zeyran

Nicknames

—	—
1	Nebula
2	Pulse
3	Vector
4	Flux
5	Zenith
6	Echo
7	Astra
8	Cipher
9	Vega
10	Helix
11	Nova
12	Phantom
13	Halo
14	Spark
15	Void
16	Prism
17	Rift

18	Shadow
19	Wraith
20	Comet

Demeanor

—	—
1	Calculating
2	Visionary
3	Stoic
4	Curious
5	Cynical
6	Resilient
7	Restless
8	Empathic
9	Reckless
10	Witty
11	Defiant
12	Mercurial
13	Pensive
14	Mysterious
15	Driven
16	Aloof
17	Idealistic
18	Pragmatic
19	Relentless
20	Inspirational

Quirks

—	—
1	Neural Implant Glitch
2	AI Best Friend
3	Cybernetic Limb Fidgeting

- | 4 | Obsessive Artifact Collector|
- | 5 | Gravitational Sensitivity|
- | 6 | Quantum Dice Gambler|
- | 7 | Nostalgic Hologamer |
- | 8 | Zero-G Anxiety |
- | 9 | Alien Language Hobbyist|
- | 10 | Tinkers with Obsolete Tech|
- | 11 | Whispering to Machines|
- | 12 | Precog Dreams |
- | 13 | Holo-Artist |
- | 14 | Collector of Alien Trinkets|
- | 15 | Genetic Augmentation Enthusiast|
- | 16 | Hates Void Silence |
- | 17 | Overly Attached to Their Drone|
- | 18 | Obsessed with Precursor Myths|
- | 19 | Experimental Augment Addict|
- | 20 | Never Removes Their Breathing Mask|

Starting Gear

Every character starts with a personal comm unit, basic vacuum suit, and 5 Supply Credits (¢) — 8¢ if playing a Merchant. Role-specific equipment is listed above. Full gear lists are in chapter 3.

Playing Synthetic and Uploaded Characters

By 4000 CE roughly 15% of core-system citizens exist as digitized minds, and many more are heavily augmented to the point where the boundary between organic and synthetic is a matter of philosophy rather than biology. Any of the ten character roles can represent a synthetic or uploaded being — use the same skills and gear, but describe them differently.

Uploaded minds typically inhabit android bodies or remote-operated frames. They do not need food, breathable air, or sleep, but they require power and are vulnerable to Cyber-Ops attacks. Their comm unit doubles as a mind-backup tether; losing it in a disaster result may mean more than lost equipment.

Synthetic intelligences that have not been human before exist as a rarer option. They often carry social stigma in frontier systems but command respect — and sometimes fear — in core polities and the Nova Terra Assembly.

Neither type has special mechanical rules. Describe the fiction and let the standard dice decide the rest.

Gear

Every character starts with a personal comm unit, basic vacuum suit, and 5 Supply Credits (¢) – 8¢ if playing a Merchant. Most standard items cost 1–2¢.

Pick additional starting gear from the lists below, subject to Referee approval. Role-specific equipment is listed in chapter 2.

Tools

- **Nanite Repair Kit:** Autonomous micro-bots for repairing equipment, vehicles, or minor structural damage.
- **Planetary Survey Gear:** Multi-spectral scanners, geologic probes, and atmosphere analyzers for mapping unknown terrains.
- **Precursor Energy Detector:** Specialized device for locating and analyzing energy signatures from ancient technologies.
- **Hacking Nodes:** Portable modules to bypass security systems or disrupt electronic networks.
- **Quantum Splicer:** A handheld tool for fine-tuning quantum circuits and repairing advanced technology.
- **Portable Fabricator:** A small, mobile 3D printer capable of creating basic tools, components, or improvised repairs.

Weapons

- **Grav-Rifle:** Fires kinetic projectiles with variable gravity fields for extra impact or ricochet effects.

- **Plasma-Edged Blade:** A close-combat weapon with a superheated plasma field for cutting through armor.
- **Disruptor Pistol:** A compact weapon designed to disable electronics, vehicles, or shields.
- **Phase Grenade:** Explosive device that creates a localized rift in space-time, disabling or scattering targets.
- **Sonic Emitter:** A non-lethal weapon emitting high-frequency waves to incapacitate opponents.
- **Void Harpoon:** A tethered, high-velocity weapon for grappling targets or traversing extreme environments.

Armor

- **Adaptive Cloak:** A stealth field generator embedded in a cloak, making the wearer blend into their surroundings.
- **Exo-Suit (with Grav-Field Dampening):** Provides enhanced strength, stability, and protection in high-gravity environments.
- **Kinetic Deflection Armor:** Absorbs and disperses kinetic impacts to protect against bullets and melee attacks.
- **Psionic Shield Harness:** Personal energy shield designed to protect against mental or psionic attacks.
- **Ablative Plating Suit:** Features layers of material designed to vaporize and absorb energy weapon impacts.

Vehicles

- **Orbital Hopper:** A compact spacecraft for short-range orbital transfers or planetary descents.
- **Grav-Bike:** A high-speed vehicle with anti-gravity stabilization for navigating uneven terrain.
- **Stellar Skimmer:** A small, agile spacecraft designed for exploration and rapid transport.
- **Mech-Assisted Rover:** A heavy-duty vehicle equipped with robotic arms and analysis tools for excavation and exploration.

- **Atmospheric Glider:** A silent, lightweight craft for scouting missions in breathable atmospheres.

Cybernetics

- **Enhanced Neural Link:** Increases reaction time and cognitive processing for faster decision-making and control of devices.
- **Hyperspectral Vision Implant:** Allows the user to see in multiple spectrums, including infrared and ultraviolet.
- **Limb with Grappling Functions:** A cybernetic arm with a retractable grappling hook for climbing or capturing.
- **Tactile Interface Augmentation:** Enhances manual dexterity for delicate operations such as repairs or medical procedures.
- **Biofeedback Regulator:** Controls stress levels and optimizes physical performance during intense activity.

Miscellaneous

- **Precursor Data Shards:** Encoded artifacts containing unknown but potentially valuable information.
- **Telepathic Relay Unit:** Enhances mental communication between users, useful for covert operations or emergencies.
- **Bio-Synth Sustenance Replicator:** Produces nutrient-rich food and water from raw organic materials.
- **Psionic Amplifier:** Boosts the user's latent psionic abilities, if any, for mental manipulation or defense.
- **Portable Cryostasis Pod:** Allows for emergency preservation of critically injured allies or valuable biological specimens.
- **Holo-Projector Kit:** Creates detailed, interactive holograms for distraction, communication, or tactical planning.
- **Atmospheric Stabilizer:** A compact device that generates breathable air in hazardous environments.

How to play

Orion: Beyond the Frontier is built for mission-based play. Each session centers on a specific job or objective with a clear scope. This keeps the system's risk-driven resolution working as intended. The rules are calibrated for situations where something is at stake, and a mission frame ensures something usually is. Extended low-pressure stretches work best as transitions between missions, not as the main mode of play.

The role of the Referee is to control the game universe: he describes situations, NPCs, and their reactions to the PCs' actions. He or she should be as clear as possible about the conditions of the context in which the characters are immersed: if there is a reasonable certainty that the character has access to a piece of information or knowledge, the referee should communicate it to the player, without asking for a roll. In this game, a roll is made only for situations of risk or uncertainty of the outcome of a direct action.

- If an action is impossible, the referee simply declares it so by explaining why and describing the possible alternatives to the player.
- If the cost of an action is explicit, the referee will clearly present it. The player can decide whether to take the action or try another one.
- If the action involves a risk, the referee will communicate this clearly. The player will be able to make informed decisions about the action they want to take and they will pull to determine the outcome.

In all other cases, the action is automatically successful.

Rolling

When a character attempts an action with significant risk, the player rolls a skill die to determine the outcome. The default is a d6.

- If the character has applicable skills or talents, they roll a bigger die like d8 or d10 to represent competence.
- Conversely, if *hindered* by injuries or obstacles may warrant a smaller die like d4 to show impairment.

The player can also roll bonus dice based on circumstances:

- Favorable conditions grant an extra d6 die. Things like having the high ground in combat or proper tools for a repair task.
- An ally who helps adds their own skill die to the roll, representing combined effort. The ally shares in any risk.

The player rolls their skill die plus any bonus dice, taking the single highest result.

- **1-2, disaster.** The full brunt of the risk occurs, and the Referee judges if there is any success at all. A roll this bad could mean instant death when mortality is on the line.
- **3-4, setback.** You suffer a lesser or partial consequence. If risking death, the character might survive but with a grievous injury and *hindered*.
- **5+, success.** The higher numbers, the better the outcomes.

Load

Characters can carry as many possessions as reasonably fits their physical capacity and the storage space on their person. For example, a character could carry a weapon, some tools, and a backpack containing rations and other useful adventuring gear.

However, if a character loads themselves down with multiple overly bulky or heavy items, it may hinder their ability to move and act freely at times. For instance, carrying a large shield, a two-handed weapon, and a bulging sack of gear may make it difficult to squeeze through narrow spaces or balance while climbing.

In such cases, the referee will inform the player that their character's encumbrance is *hindering* their movement or dexterity in the current situation. The player can then decide to temporarily set down some of their carried items to proceed unfettered, stowing the extra gear nearby to be retrieved later. Or they may opt to continue carrying everything, accepting additional challenges or risks to their intended actions.

Advancement

After successfully completing a mission or major story milestone, each player character will improve in the following ways as a reward for their accomplishments:

- Increase one of their skill ranks by one step on the progression track (none → d8 → d10 → d12). This represents honing their competencies through experience.
- Gain a number of currency credits (¤) equal to a d6 roll. Credits can be spent to acquire new gear, weapons, cyberware, services, etc. The die roll result encourages uncertainty in the exact amount of new wealth obtained.

For example, after infiltrating an enemy base undetected and stealing valuable data, a character with stealth and hacking skills currently at d8 could increase their stealth to d10 for greater sneaking ability. They also would gain somewhere between 1-6 credits to spend on desired upgrades and purchases.

Damage Control

Say how you use a piece of gear to absorb damage. It's disabled until repaired at a base.

When a character is about to take damage from an attack or hazard, the player can describe how they utilize a piece of equipment to absorb some or all of the incoming damage. This could involve actions like raising a shield to block an incoming blow, using a large tool as cover from an explosion, or angling a vehicle to have an attack hit a reinforced section.

After being used to soak damage this way, that gear becomes disabled and non-functional until it can be repaired back at a home base or facility with the proper tools and parts.

The referee will determine if using the gear fully negates the attack or just reduces the damage taken. More powerful or intense attacks may overwhelm a piece of gear's protection capacity.

Injuries

If a character sustains a serious injury during a game session, they will need to spend some in-game time in the medical bay to recover. The player should describe their character receiving medical treatment and the referee can narrate the passage of time during the healing process.

If a character is killed, the player should introduce a new character as soon as is practical. The referee should facilitate quickly integrating the new character into the ongoing story.

When adjudicating the effects of injuries or character death, inclusion of the affected player takes priority over adhering to strict realism. The referee should aim to avoid excluding a player from participation due to their character being incapacitated. Workarounds like having the player take temporary control of a supporting character can help keep everyone involved.

The risks of combat and other dangerous activities are an important part of the game's drama and stakes. However, the ultimate goal is for everyone to have fun together. So when serious consequences occur, the referee should guide the narrative back toward the affected player's active participation.

Game mastering

The referee portrays non-player characters (NPCs) by describing their behaviors, motivations, and obstacles they present to the player characters. The referee should lead the group in establishing boundaries for subject matter and conduct during play.

To control pacing and safety, the referee can fast-forward through uneventful stretches, pause play for breaks or sidebar discussions, or even rewind and redo a scene if needed. The referee should invite players to request scene edits too.

The referee should present dilemmas and challenges without pre-planned solutions, letting players direct the story.

To give everyone screen time, the referee checks in with each player in turn.

When careful play has kept the dice quiet for a while, test for bad luck: roll a die — 1–2 means trouble surfaces, 3–4 means signs of trouble ahead. Before a session of low-profile play, name at least one ambient pressure that fits the mission: fuel shortage, time pressure, missing clearance, equipment wear, communications delay, political scrutiny. These conditions give the bad luck check something concrete to trigger.

When the written rules don't adequately cover a situation, the referee improvises rulings in the moment to keep the game moving. If any rulings seem unsatisfactory, the group should discuss them on a break and revise them collaboratively.

Situation

When characters act without risk, the GM rolls a die sized to how settled things are: d4 (chaotic), d6 (precarious), d8 (manageable), d10 (predictable), d12 (stable), to find how the world moves.

- 1–2 Trouble. An adverse fact surfaces, something the characters can see and respond to.
- 3–4 Signs. Trouble shows at the edge; the situation grows less stable, the next die will likely be smaller.
- 5+ Calm. Nothing new; the action lands.

The surfaced fact must be proportionate to what is established, leave something to act on, and follow from prior events, not invented to fill the silence.

Opponents

Characters played by the GM have no skill dice. Define them by their **behavior** (what they want, including when they would rather flee or bargain than fight), the **risks** they present (the 1–2 result for players in conflict with them), and any **obstacles** (visible conditions, armor, numbers, cover, that must be cleared before they can be put down). Signal both before players roll.

Conflict

In a fight, the GM rolls the situation die: d4 if ambushed or scattered, d6 if evenly matched, d8 if holding an advantage, d10 if dominating, d12 if the opposition is broken, to find what the opposition does.

- 1–2 Press. A new risk or obstacle surfaces; the fight grows harder.
- 3–4 Shift. The opposition adapts, repositions, changes tactic; the situation tilts.
- 5+ Opening. An obstacle clears, a gap appears, or the opposition is ready to break.

Roll in lulls, at the start of a fight, or whenever no skill roll applies. For extended conflicts, use Steps: state actions and risks, roll as needed, describe how the situation has changed, repeat. On an Opening with the opposition cornered, they break per their behavior, fleeing, yielding, or bargaining.

Large-Scale Threats

When a mission involves the Varnon Swarm, a Precursor site, or another phenomenon beyond ordinary human control, treat it as a source of pressure rather than a separate system. Use the existing mission frame, hazard tables, and consequences to show the threat advancing or changing shape. If the crew learns something that should not be public, or their actions reveal a larger pattern, let that knowledge reshape future missions rather than resolving the threat immediately.

Playing Solo

Orion: Beyond the Frontier works as a solo game. You play your character; two oracles cover everything else.

The **Risk Oracle** tells you how much control you have before you act — whether the situation is stable enough to skip a roll or volatile enough to make failure hit hard. The **Question Oracle** answers yes/no questions about the world: NPC reactions, off-screen events, things your character can't directly observe. Use the Sparks tables when you need a fresh direction and nothing comes to mind.

You don't need to use all three tools every scene. Most moments resolve through the Risk Oracle alone. The Question Oracle comes in when the fiction raises a question your character can't answer by acting. Sparks are a last resort, not a crutch.

The Loop

Each time your character acts, go through these steps:

1. **Declare your action.** What are you doing, and what do you want to happen?
2. **Assess the situation.** Use the *Situation Roll* to read how much is at stake.
3. **Roll if needed.** If the Situation Roll says there's friction or volatility, make your standard roll. Otherwise, it just happens.
4. **Interpret the outcome.** Let the result shape the fiction. If you need to know how the world or an NPC responds, use the *Question Oracle*.

Question Oracle

When the outcome isn't about what your character *does* — NPC reactions, world states, things off-screen — ask a yes/no question and roll based on the odds.

Likelihood	Die
Very Unlikely	d4
Unlikely	d6
Likely	d8
Very Likely	d10

Likelihood	Die
Almost Certain	d12

Interpret the result:

- **1-2 — No, and...** It doesn't happen, and things get worse.
- **3-4 — Yes, but...** It happens, but with a complication.
- **5+ — Yes, and...** It happens, and something else opens up.

Sparks

When you need a fresh angle or you're stuck, combine an Action and a Theme to generate a prompt.

Action

D20	Action
1	Investigate
2	Rescue
3	Sabotage
4	Pursue
5	Retrieve
6	Expose
7	Defend
8	Infiltrate
9	Negotiate
10	Escape
11	Salvage
12	Decrypt
13	Protect
14	Surveil
15	Confront
16	Transport

D20	Action
17	Warn
18	Steal
19	Betray
20	Liberate

Theme

D20	Theme
1	Ruin
2	Jumpgate
3	Resonant
4	Megastructure
5	Shard
6	Eclipse
7	Varnon
8	Rivalry
9	Upload
10	Contract
11	Zynthari
12	Anomaly
13	Colony
14	Alien
15	Contraband
16	Transhuman
17	Scar
18	Tevkar
19	Void
20	Derelect

Interstellar Travel

Interstellar travel in *Orion: Beyond the Frontier* is a test of resource management, advanced technology, and tactical decision-making at every jump. Humanity relies on two principal technologies: the relatively new **Phase Resonance Drives** and the legacy network of **Jumpgates**, each with its own constraints and politics. Travel across the Orion Frontier is fraught with unknowns: Precursor relics that warp space-time, alien patrols guarding restricted zones, and anomalies that defy physics.

Navigation Hazards

Navigating the Orion Frontier requires precision, adaptability, and vigilance. The vastness of space harbors countless dangers that can disrupt even the most well-planned routes. Skilled characters and advanced technology are critical to surviving the perils of interstellar travel.

During journeys, travelers may encounter:

1. **Astrogation Challenges:** Mapping courses through Phase Resonance corridors or avoiding unstable jumpgate remnants requires advanced calculations. Navigation checks determine efficiency and avoid costly delays or resource depletion.
2. **Void Lattice Disturbances:** Unstable areas of space, often tied to Precursor technology, create rifts or localized gravity distortions. These zones are unpredictable and may cause malfunctions or displace ships into uncharted regions.

3. **Radiation Storms:** Areas of intense radiation can disable systems or harm crew. Effective shielding, evasive maneuvers, or detours may be necessary to avoid catastrophe.
4. **Gravitational Anomalies:** Rogue black holes or collapsing neutron stars can warp space-time, trapping unprepared ships. Advanced sensors and expert piloting are essential to escape these zones.
5. **Alien Patrol Zones:** Restricted sectors, particularly near Zynthari or independent rogue AI territories, are heavily monitored. Trespassing may provoke hostile responses, requiring diplomacy or combat readiness.
6. **Celestial Hazards:** Massive asteroid fields, rogue comets, and stellar debris from ancient battles pose navigational challenges. Evasive maneuvers and repair capabilities are vital.
7. **Subspace Phenomena:** Unstable pockets of space can destabilize Phase Resonance Drives, stranding ships far from their intended destination.

Core Technologies for Travel

1. **Phase Resonance Drives:** The primary FTL method of 4000 CE. PRDs resonate with the fabric of space-time to enable near-instantaneous transit between calibrated resonance points, with a maximum range of 30 light-years per jump. They are prone to destabilization near void rifts or dense Precursor anomalies, and expanding the network requires locating and calibrating new resonance points — a task that draws explorers into uncharted space. PRD nodes are often discovered near ancient Precursor ruins.
2. **Jumpgates:** Legacy infrastructure from the interstellar expansion of the 2100s–2400s, now rare and increasingly faction-controlled. Those that survive provide near-instant transit between connected systems up to 15 light-years apart, but operating them requires tolls, faction authorization, or hacking. Some gates have fallen into disuse or been seized by the Sirius Compact or IFMC as strategic choke points. Where PRDs are the open road, jumpgates are the toll bridges.

Skills for Interstellar Success

Horizon Pilots, Tech-Scavengers, and Archivists are the roles best suited to managing the demands of interstellar travel.

Random Interstellar Encounters

Roll 1d12 to determine a random encounter in the unpredictable and mysterious **Orion Frontier**:

1. **Void Rift Anomaly:** The ship encounters a pocket of unstable space-time, warping sensors and navigation systems. Emergency adjustments are needed to avoid being displaced into an unknown region.
2. **Distress Beacon:** A distress signal from a stranded or damaged vessel invites investigation. It could lead to survivors, valuable resources, or an ambush.
3. **Ancient Debris Field:** The ship drifts into the remnants of a Precursor battle or megastructure. Navigating the wreckage offers risks and opportunities for salvage.
4. **Derelict Precursor Vessel:** A dormant alien craft of enigmatic design floats in the void. Boarding it may reveal advanced technology or trigger unexpected defenses.
5. **Astrogration Malfunction:** A systems failure leaves the crew relying on manual navigation to avoid drifting off-course. This demands quick thinking and precise calculations.
6. **Gravitational Drift:** A sudden encounter with a rogue black hole or massive gravity well pulls the ship off-course. Evasive maneuvers are required to escape its grasp.
7. **Subspace Corridor Instability:** A shortcut through a subspace tunnel opens, but entering it carries risks of disorientation, time dilation, or system-wide failure.
8. **Rogue Faction Ambush:** A hostile faction, such as space pirates or rogue AI-controlled ships, attempts to disable or board the vessel. Prepare for combat or negotiation.

9. **Interstellar Trade Convoy:** A group of independent traders or smugglers offers goods, intel, or barter. They may also share warnings about local dangers—or have hidden motives.
10. **Celestial Phenomenon:** The ship encounters a rare cosmic event, like a quasar pulse or an exploding star. It could be a scientific breakthrough or a threat to nearby systems.
11. **Drifting Artifact:** Sensors detect a mysterious alien object emitting faint energy signals. Its true nature is unknown; it could be valuable, dangerous, or both.
12. **AI Outpost Contact:** The ship receives a cryptic message from a rogue AI installation. The AI might offer trade, riddles, or a trap.

Travel Time and Fuel Consumption

Two principal methods of travel exist: **conventional interplanetary propulsion** and **Phase Resonance technology**, a successor to jumpgates. Both demand planning; neither is cheap.

Interplanetary Travel

Voyages within star systems rely on conventional propulsion, updated with advanced materials and efficiency.

- **Fusion Torches and Ion Drives:** Still the workhorses of interplanetary travel, providing sustained acceleration for cargo, personnel, and exploration missions.
- **Phase Resonance Boosters:** These augment sublight speeds for quicker traversal within systems, though at higher energy costs.

Travel times between planets, while reduced, still take days to weeks depending on distance. For example:

- A journey equivalent to Earth to Neptune takes **4-6 weeks** using modern fusion engines.
- Resource efficiency is improved, but heavy loads such as colonization equipment still demand significant fuel resources.

Fuel Requirements:

Fuel use depends on the ship's mass and energy requirements:

- **Small Craft (Exploration Vessels):** Light fuel consumption, suitable for extended missions with occasional refueling.
- **Heavy Cargo Haulers:** Require large fusion cores or antimatter reserves for longer trips.

Key Considerations:

- **Reliability:** Fusion and ion drives are dependable but slow.
- **Resource Extraction:** In-system mining remains a critical aspect of maintaining fuel reserves.
- **Acceleration Limits:** Pushing for maximum speed requires exponentially more fuel and increases risks of mechanical stress.

Phase Resonance Technology

Phase Resonance Drives (PRDs) allow near-instantaneous transit between aligned star systems. Using spacetime resonance fields derived from Precursor-artifact research, these drives bypass light-speed limitations. They are the dominant FTL method but are not without constraints:

Key Features:

- **Aligned Systems:** Only systems with known and calibrated resonance points are accessible. Discovering and calibrating new nodes is a primary driver of frontier exploration.
- **Instantaneous Transit:** Travel between aligned systems takes seconds, but initiating the drive consumes enormous energy.
- **Resource Costs:** PRDs require antimatter and rare materials, limiting their use to major factions and well-funded explorers.

Limitations:

- **Range:** Maximum effective range between resonance points is **30 light-years**, double that of old jumpgates, but still restrictive compared to galactic scales.

- **Risk of Malfunction:** Improper calibration can result in catastrophic outcomes, including dimensional rift creation or complete vessel destruction.

Hybrid Travel

The combination of **interplanetary propulsion** and **Phase Resonance Drives** defines most voyages in the Orion sector:

- **In-System Maneuvers:** Fusion engines navigate local celestial hazards and bring vessels into alignment with resonance points.
- **Interstellar Jumps:** Once aligned, PRDs allow rapid leaps to distant systems, enabling trade and exploration across far-flung colonies.

Key Travel Strategies

1. **Careful Route Planning:** PRD-capable ships must carefully plot trajectories to align with resonance points while conserving antimatter reserves for jumps.
2. **Refueling Operations:** Colonies and space stations are essential for resource replenishment. Frontier explorers often rely on asteroid mining or alien energy sources.
3. **Avoiding Hazards:** Navigating cosmic phenomena like void rifts, rogue planets, or pirate zones adds layers of risk.

The Future of Travel

Rumors persist that the Phase Resonance network is only a surface layer of a deeper Precursor architecture — a fully integrated interstellar web that would allow transit without calibration costs or antimatter expenditure. No faction has proven this, but it drives much of the research conducted near ancient ruins. If the rumor is true, humanity has been rediscovering a road the Precursors paved millions of years ago.

Environmental Hazards

Environmental hazards in *Orion: Beyond the Frontier* highlight the harsh realities of exploring and surviving in the uncharted reaches of the galaxy. From unpredictable anomalies to resource scarcity and extreme conditions,

these challenges test characters' adaptability and decision-making skills. Encounters with these hazards are integral to the game's survival mechanics, underscoring the fragility of life in the vastness of space.

Radiation Exposure

Space is saturated with dangerous radiation—from stellar flares to bursts from nearby quasars or unstable Precursor relics.

- **Navigating Radiation Zones:** Characters must carefully chart routes to avoid high-radiation areas or rely on specialized shielding.
- **Shielding and Monitoring:** Equipment like radiation suits, ship shielding upgrades, and dosimeters are vital.
- **Mechanics:** Exposure to radiation could result in cumulative health effects or equipment malfunctions. Players may face critical decisions, such as prioritizing repairs or taking detours to avoid radiation-heavy regions.

Microgravity and Variable Gravity

In the Orion sector, microgravity conditions are only part of the challenge—some zones feature unstable or extreme gravity fluctuations due to Precursor artifacts or natural phenomena.

- **Microgravity Effects:** Weightlessness impacts physical movement, coordination, and even mental acuity over time.
- **High-Gravity Zones:** Prolonged exposure to intense gravity fields can strain equipment and exhaust characters.
- **Specialized Tools:** Magnetic boots, handheld propulsion devices, or gravity stabilizers are often essential.
- **Mechanics:** Characters may need to succeed at gravity-adaptation checks to perform tasks or risk penalties such as slower action times or physical strain.

Extreme Temperatures

Space exploration involves exposure to deadly extremes of heat and cold, from the blazing proximity of a star to icy voids on the edge of systems.

- **Temperature Challenges:** Planets, derelict stations, or rift anomalies may exhibit wild temperature swings requiring constant monitoring.
- **Gear Requirements:** Thermal suits, adaptive environmental systems, and vehicle insulation are mandatory.
- **Mechanics:** Failure to prepare for temperature extremes can damage systems, cause hypothermia or heat stroke, and impact performance. Characters must make temperature regulation checks or face escalating consequences.

Scarcity of Resources

Frontier exploration is inherently resource-starved. Supplies such as oxygen, food, water, and energy are finite and must be managed carefully.

- **Resource Management:** Players need to balance exploration priorities with the availability of consumables. Mismanagement can lead to perilous shortages.
- **Acquisition Missions:** Encounters may involve scavenging wreckage, mining asteroids, or negotiating with rival factions for critical supplies.
- **Mechanics:** Scarcity can create high-stakes tension, with characters forced to decide between extending their reach or retreating to resupply.

Spatial and Anomalous Hazards

The Orion sector harbors spatial phenomena that challenge conventional physics and endanger ships and crews.

- **Spatial Rifts:** Unstable anomalies can distort navigation, cause temporal disorientation, or draw vessels into unpredictable regions of space.
- **Energy Fluctuations:** Massive bursts from alien devices or natural phenomena can overload systems and disable ships.
- **Debris Clouds:** Areas filled with micrometeoroids or ancient battle remnants may tear through unshielded vessels.
- **Mechanics:** These hazards require high-stakes navigation rolls or creative problem-solving to overcome.

Environmental Suits and Scientific Equipment

Explorers in the Orion Frontier rely on highly advanced suits and tools to survive hostile environments and gather critical data.

- **Environmental Suits:** Modular, adaptive suits offer protection against extreme conditions, radiation, and atmospheric toxins.
- **Scientific Gear:** Portable analyzers, xenoscopes, and bio-scanners allow players to investigate alien phenomena, study ecosystems, and locate valuable resources.
- **Mechanics:** Equipment failure in critical moments can force players to improvise solutions or risk mission failure.

Vehicles of the Orion Frontier

Technological advancement over millennia has drastically transformed vehicles into highly specialized and adaptable tools for interstellar exploration, colonization, and resource extraction. Here's an updated catalog of vehicles, reflecting the far-reaching technological progress of 4000 CE in the Orion Frontier.

Phase-Jump Shuttle

A next-generation transport craft for rapid interplanetary and short-range interstellar travel.

- **Features:** Equipped with **Phase Resonance Nodes**, enabling near-instantaneous jumps within a system or to nearby systems.
- **Usage:** Primarily used for personnel transfer, supply runs, or emergency evacuations. Its self-contained life-support pods allow independent operations for months if stranded.
- **Unique Tech:** Adaptive shield generators protect against subspace distortions during transit.

Terraformer Rover

A mobile, multi-purpose vehicle designed to prepare planetary surfaces for colonization.

- **Features:** Includes **atmospheric processors**, **hydrocarbon scrubbers**, and autonomous drone networks for large-scale terrain alteration.
- **Usage:** Deployed on barren or inhospitable worlds to shape landforms, establish basic infrastructure, and seed atmosphere stabilizers.
- **Unique Tech:** Self-repairing hull with nanite integration allows it to function in extreme environments for decades.

Cryo-Vault Terrain Skimmer

An advanced craft for operations on cryogenic worlds where temperatures reach near-absolute zero.

- **Features:** Incorporates **quantum-insulated plating**, advanced thermal management systems, and plasma jet propulsion.
- **Usage:** Ideal for mining frozen moons, analyzing ice-world ecosystems, or recovering volatile compounds from icy comets.
- **Unique Tech:** Adaptive plasma blades cut through frozen terrain to extract subsurface resources.

All-Terrain Exo-Strider

A versatile exploration vehicle resembling a bipedal walker.

- **Features:** Articulated legs with **gravitic stabilizers** for traversing jagged cliffs, lava fields, or dense jungles.
- **Usage:** Operates in uncharted or high-risk regions where traditional rovers fail. Often paired with scout drones.
- **Unique Tech:** **Bio-reactive plating** that can alter its surface to blend into various terrains, providing stealth capabilities.

Atmospheric Phantom Skimmer

An ultra-fast, low-altitude flyer for reconnaissance and rapid transport across planetary atmospheres.

- **Features:** Silent anti-gravity drives and a cloaking field, making it ideal for stealth missions.
- **Usage:** Used by explorers, diplomats, or reconnaissance teams in contested or mysterious regions.

- **Unique Tech: Aero-luminal sensors** map atmospheric anomalies and detect alien lifeforms at high speeds.

Stellar Nexus Freighter

An enormous cargo vessel equipped for interstellar trade and transport.

- **Features:** Houses **quantum-tunneling storage bays** that allow massive capacity while minimizing spatial footprint.
- **Usage:** Facilitates the movement of materials, goods, and colonization equipment between distant star systems.
- **Unique Tech: Temporal stasis fields** preserve perishable goods or fragile materials for long journeys.

Gas-Harvesting Leviathan

A colossal extractor ship designed to mine gas giants and dense atmospheric layers.

- **Features:** Massive, flexible scoop systems, **molecular compressors**, and integrated refineries process gases into usable fuels or exotic elements.
- **Usage:** Supports resource collection for advanced propulsion systems or industrial production in core systems.
- **Unique Tech: Gravity dampeners** stabilize operations in turbulent atmospheric currents.

Void-Crawler Scavenger

A specialized salvage craft for deep-space operations.

- **Features:** Equipped with **magnetic grapplers**, **debris processors**, and precision-cutting lasers for dismantling derelict stations, ships, or Precursor ruins.
- **Usage:** Often employed by scavenger crews exploring uncharted systems for artifacts or rare materials.
- **Unique Tech: AI-assisted analysis suites** identify valuable components or analyze alien technology on the fly.

Hyper-Adaptive Explorer Sphere

A modular, self-contained exploration craft capable of adapting to any environment.

- **Features:** Spherical design with **shape-memory outer shells** for extreme flexibility in hostile or unpredictable conditions.
- **Usage:** Used by scientific expeditions to study anomalies, dimensional rifts, or newly discovered planets.
- **Unique Tech:** **Dimensional phase-shifters** allow temporary entry into alternate spatial layers for unique research opportunities.

Precursor-Adapted Scout Pod

A compact exploration craft reverse-engineered from Precursor ruin components grafted onto a human-built frame.

- **Features:** Hull sections of unknown alloy composition resist radiation and gravitational distortion far better than manufactured equivalents. Equipped with a **Precursor Spatial Awareness Array** that passively maps nearby dimensional anomalies and Phase Resonance points.
- **Usage:** Favored by Precursor Seekers and tech-scavengers entering void rifts or unstable megastructure interiors. The navigational interface requires Sensors to use effectively; without it the craft is flyable but blind to its most valuable data streams.
- **Unique Tech:** The alien hull geometry grants a natural stealth profile against conventional sensors, though the reason remains poorly understood.

Starships

Starships are equipped with advanced versions of the following key functions. Further upgrades are available at specialized shipyards or Precursor hubs. In critical situations, the crew can perform specific actions to adapt or optimize their systems.

COMMS:

- **Standard:** Quantum-encoded communication systems with basic inter-stellar range.
- **Upgrades:** Hyperwave transceivers, AI-assisted negotiation relays.

EMBARKATION:

- **Standard:** Modular personnel shuttles and emergency escape pods.
- **Upgrades:** Orbital dropships, dimensional boarding probes.

HULL:

- **Standard:** Adaptive nano-alloy with embedded self-repair systems.
- **Upgrades:** Ablative phase armor, Precursor material plating.

LIFE SUPPORT:

- **Standard:** Advanced atmospheric scrubbers and nutrient synthesizers.
- **Upgrades:** Cryonic preservation chambers, bio-integration systems.

SENSORS:

- **Standard:** Multi-spectrum scanners for navigation and basic detection.
- **Upgrades:** Gravimetric resonance arrays, temporal echo scanners.

POWER:

- **Standard:** Quantum fusion reactor with redundant energy grids.
- **Upgrades:** Antimatter containment units, dark energy converters.

PROPULSION:

- **Standard:** Ion-plasma drives with gravitational stabilizers.
- **Upgrades:** Phase resonance engines, subspace inversion thrusters.

DEFENSES:

- **Standard:** Standard plating with integrated energy deflection fields.
- **Upgrades:** Quantum shield generators, void interceptor drones.

ARMAMENTS:

- **Standard:** None.
- **Upgrades:** Precursor energy lances, phase disruption torpedoes, kinetic rail cannons.

Wise captains balance exploration, combat readiness, and resource efficiency. The right upgrades determine what a ship can handle and what it cannot.

Starship Types and Classes

Below are the primary types and classes of starships suited to the setting of 4000 CE.

Interstellar Explorers

Built for long-range exploration and survey, often operating far from resupply.

- **Pioneer Vessel:** Massive ships equipped for multi-decade voyages, combining advanced laboratories, extensive crew quarters, and self-sustaining systems. Perfect for deep exploration into uncharted regions.
- **Pathfinder:** Smaller, faster exploration vessels specializing in charting Phase Resonance points and surveying potentially habitable systems.
- **Xenobiology Platform:** Focused on studying alien ecosystems, these vessels house bio-containment labs and environmental simulators for analyzing extraterrestrial life.
- **Starprobe Carrier:** Unmanned drone carriers capable of deploying swarms of autonomous probes to map hazardous regions or Precursor ruins from a safe distance.

Scientific Ships

Craft dedicated to understanding and utilizing the scientific mysteries of the universe.

- **Temporal Research Vessel:** Equipped with temporal scanners and containment systems to study time-based anomalies and echoes of past events.
- **Quantum Analytics Cruiser:** Focused on subatomic studies, these ships refine dark energy and antimatter production techniques.
- **Precursor Artifact Transporter:** Specially designed to safely secure and study unstable alien relics, featuring containment fields and dedicated resonance dampeners.

Colony Ships

Vessels built to establish or support human settlements on distant worlds.

- **Interstellar Ark:** Gigantic ships transporting entire populations, modular habitats, and terraforming equipment to uninhabited planets.
- **Terraformer Cruiser:** Deploys atmospheric processors, hydrocarbon scrubbers, and robotic terraforming units to prepare planets for long-term habitation.
- **Colonial Vanguard:** Smaller than Arks, these ships carry pre-fabricated structures, resource processors, and initial colonists to lay the groundwork for future expansions.

Cargo and Mining Ships

Focused on resource extraction and transportation across vast distances.

- **System Hauler:** Modular freighters optimized for bulk transport of goods, materials, and colony supplies between systems.
- **Asteroid Miner:** Ships equipped with gravimetric drills and ore processing facilities for large-scale asteroid mining operations.
- **Gas Harvester:** Designed to skim gas giant atmospheres, extracting exotic gases and converting them into usable resources.

Combat and Defense Ships

Military-focused starships tasked with safeguarding humanity from external threats.

- **Stellar Defender:** Medium-sized ships outfitted with energy weapons and interceptor drones, designed to protect trade routes and colonies.
- **Dreadnought:** Heavily armored warships bristling with railguns, energy lances, and phase disruptors, built for major engagements.
- **Void Interceptor:** Agile fighters equipped with advanced cloaking devices, perfect for rapid strikes or defensive maneuvers.
- **Guardian Frigate:** Multi-role defensive ships designed to patrol planetary systems and engage smaller threats.

Trade and Transport Ships

Facilitators of commerce and diplomacy, ensuring the movement of goods and people.

- **Stellar Merchantman:** Hybrid cargo-passenger ships designed for long-haul trade and interstellar commerce.
- **Diplomatic Envoy Ship:** Equipped with advanced communication systems and luxurious accommodations, these ships foster negotiations and alliances.
- **Quantum Courier:** Rapid transport vessels for high-priority goods and personnel, featuring advanced slipstream drives for speed.

Scout and Reconnaissance Ships

Designed for intelligence gathering and advanced warning in hostile or unknown regions.

- **Void Scout:** Compact vessels equipped with hypersensitive sensors and gravitic cloaking, ideal for deep reconnaissance missions.
- **Anomaly Mapper:** Specializes in charting dangerous regions, like rift fields or radiation zones, providing data for larger fleets.
- **Phase Boundary Surveyor:** Tailored to explore unstable resonance points and calibrate new paths for interstellar travel.

Medical and Rescue Ships

Dedicated to providing care and responding to crises across the frontier.

- **Stellar Hospital:** Large, state-of-the-art medical ships offering advanced surgical suites, bio-engineering labs, and trauma facilities.
- **Rescue Frigate:** Quick-response ships designed to extract survivors from hostile environments or damaged vessels.
- **Cryostasis Support Ship:** Provides long-term preservation for critically injured individuals or valuable biological samples.

Specialized Precursor Technology Ships

Advanced craft leveraging mysterious alien technology.

- **Artifact Harvester:** Ships equipped with resonance shields and precision manipulators to recover fragile alien artifacts.
- **Dimensional Rift Navigator:** Tailored to traverse unstable rifts and explore Precursor megastructures, these ships blend human and alien tech.
- **Starforge Cruiser:** Rare, experimental ships capable of assembling or repairing large structures using Precursor-derived manufacturing systems.

Hybrid Ships

Vessels designed for multi-role functionality, blurring traditional categories.

- **Exploration and Combat Hybrid (ECHO):** Combines high-end survey equipment with advanced weaponry, ideal for exploring contested systems.
- **Colony Combat Support (CCS):** Ships designed to establish settlements in hostile environments, equipped with defensive capabilities.
- **Trader-Explorer:** Built for independent ventures, offering storage capacity, exploration tools, and basic defenses for frontier entrepreneurs.

Combat

Conflict Resolution

If conflicts arise between factions during interplanetary travel, space combat mechanics can be used. Combat follows a similar structure to regular combat, with rounds and action sequences.

Ship Attributes

Spacecraft have their own functions, such as Hull, Weapons, Armor, and Propulsion. These determine the capabilities and effectiveness of the spacecraft in combat situations. The Referee can create a set of ship attributes or use pre-defined ship templates for ease of play.

Ship Actions

During space combat, characters can take various ship actions, such as attacking with weapons, evasive maneuvers, deploying countermeasures, or targeting specific ship systems. The actions available depend on the ship's capabilities and the character's skills.

Combat Rolls

Combat rolls are used to determine the outcomes of attacks and defensive actions. These rolls take into account the attacker's skill, ship attributes, and potential modifiers based on the situation. Damage inflicted on the target ship is calculated, considering the defender's armor and other defensive systems.

Boarding Actions

In addition to ship-to-ship combat, characters may engage in boarding actions, attempting to breach enemy ships and engage in close-quarters combat. Boarding actions follow similar rules to regular combat, with characters utilizing their skills and weapons to overcome enemy forces.

Consequences and Repairs

Space combat can have long-lasting consequences, damaging ship systems and potentially causing critical failures. Characters must deal with the

aftermath of combat, including repairing their ships, tending to injured crew members, and managing limited resources in the hostile environment of space.

Shipyard

This section presents sample ships from the diverse classes of starships operating in the Orion Frontier. From deep-space exploration to planetary defense, each vessel is equipped to meet the challenges of its role while serving as a lifeline for its crew in the vast expanse of space.

Interstellar Explorers

Pioneer Vessel: *Horizon Eternal*

Large, versatile ships built for prolonged interstellar missions, combining exploration tools with a self-sustaining habitat.

- **COMMS:** Quantum Hyperwave Communicator
- **EMBARKATION:** Modular Drop Shuttles
- **HULL:** Adaptive Nano-Alloy
- **LIFE SUPPORT:** Integrated Bio-Systems
- **SENSORS:** Gravimetric Resonance Arrays
- **POWER:** Quantum Fusion Reactor
- **PROPULSION:** Phase Resonance Engines
- **DEFENSES:** Ablative Phase Armor
- **ARMAMENTS:** Precursor Energy Lances

Pathfinder: *Horizon Seeker*

A swift and compact exploration ship optimized for locating resonance points and charting uncharted territories.

- **COMMS:** Multi-Spectrum Transmitter
- **EMBARKATION:** Light Recon Pods
- **HULL:** Lightweight Composite Frame
- **LIFE SUPPORT:** Nutrient Synth Replicators
- **SENSORS:** Phase Harmonic Scanners

- **POWER:** Mini Fusion Core
- **PROPULSION:** High-Efficiency Plasma Drives
- **DEFENSES:** Energy Deflection Fields
- **ARMAMENTS:** Point-Defense Lasers

Xenobiology Platform: *Biofinder Nova*

A floating laboratory designed to investigate and study alien ecosystems and microbial life.

- **COMMS:** AI-Assisted Data Relays
- **EMBARKATION:** Containment Drones
- **HULL:** Reinforced Bio-Shielding
- **LIFE SUPPORT:** Contaminant Isolation Chambers
- **SENSORS:** Biochemical Spectroscopes
- **POWER:** Antimatter Energy Cell
- **PROPULSION:** Ion-Plasma Thrusters
- **DEFENSES:** Bioadaptive Energy Shields
- **ARMAMENTS:** None

Starprobe Carrier: *Vanguard Harbinger*

Designed to deploy and coordinate autonomous probe swarms for safe mapping of hazardous regions.

- **COMMS:** Probe Control Network
- **EMBARKATION:** Automated Probe Bays
- **HULL:** Reinforced Alloy Shell
- **LIFE SUPPORT:** Basic Long-Range Systems
- **SENSORS:** Quantum Scan Arrays
- **POWER:** Dual Fusion Generators
- **PROPULSION:** Grav-Optimized Drives
- **DEFENSES:** Void Disruption Flares
- **ARMAMENTS:** Railgun Batteries

Scientific Ships

Temporal Research Vessel: *Chronos Insight*

Dedicated to studying temporal anomalies and Precursor remnants, equipped with state-of-the-art containment systems.

- **COMMS:** Temporal Distortion Communicator
- **EMBARKATION:** Modular Containment Pods
- **HULL:** Temporal-Resistant Alloy
- **LIFE SUPPORT:** Advanced Environmental Control
- **SENSORS:** Temporal Wave Monitors
- **POWER:** Dark Energy Generator
- **PROPULSION:** Subspace Inversion Thrusters
- **DEFENSES:** Time-Lock Barriers
- **ARMAMENTS:** Temporal Disruption Torpedoes

Quantum Analytics Cruiser: *Eidolon Prism*

A mobile lab specializing in quantum mechanics and dark energy research.

- **COMMS:** Subatomic Data Transceivers
- **EMBARKATION:** Enclosed Drone Bays
- **HULL:** Quantum-Fused Plating
- **LIFE SUPPORT:** Cryogenic Support Systems
- **SENSORS:** Subspace Particle Arrays
- **POWER:** Antimatter Core
- **PROPULSION:** Gravimetric Warp Coils
- **DEFENSES:** Particle Deflection Fields
- **ARMAMENTS:** None

Colony Ships

Interstellar Ark: *Genesis Ascendant*

A massive ship carrying thousands of colonists and modular infrastructure for planetary settlement.

- **COMMS:** Stellar Uplink Array

- **EMBARKATION:** Habitat Deployment Systems
- **HULL:** Self-Healing Nano-Mesh
- **LIFE SUPPORT:** Closed-Loop Ecosystem
- **SENSORS:** Planetary Survey Suites
- **POWER:** Dyson Micro-Forge Core
- **PROPULSION:** Phase Resonance Engines
- **DEFENSES:** Orbital Defense Drones
- **ARMAMENTS:** Minimal

Combat and Defense Ships

Stellar Defender: *Aegis Vanguard*

A medium-sized ship equipped to defend trade routes and colonies with advanced weaponry.

- **COMMS:** Fleet Command Module
- **EMBARKATION:** Interceptor Launch Bays
- **HULL:** Reinforced Carbon-Plasteel Armor
- **LIFE SUPPORT:** Military-Grade Filters
- **SENSORS:** Combat Range Enhancers
- **POWER:** Antimatter Overclock Engine
- **PROPULSION:** Dual Ion Drives
- **DEFENSES:** Quantum Shields
- **ARMAMENTS:** Phase Lances, Missile Pods

Dreadnought: *Eternal Bastion*

An intimidating warship designed for large-scale engagements and heavy firepower.

- **COMMS:** Battle Network Relay
- **EMBARKATION:** Heavy Assault Pods
- **HULL:** Precursor-Infused Armor
- **LIFE SUPPORT:** Full Crew Complement
- **SENSORS:** Enhanced Tactical Arrays

- **POWER:** Planetary Core Reactor
- **PROPULSION:** Quantum Turbine Drives
- **DEFENSES:** Reactive Plating, Shield Webs
- **ARMAMENTS:** Railgun Cannons, Disruption Torpedoes

Scout and Reconnaissance Ships

Void Scout: *Phantom Runner*

A stealthy, compact vessel perfect for gathering intelligence in hostile regions.

- **COMMS:** Silent Signal Relay
- **EMBARKATION:** Recon Drone Launchers
- **HULL:** Adaptive Cloaking Surface
- **LIFE SUPPORT:** Minimalistic Survival Pods
- **SENSORS:** Advanced Spectrum Scanners
- **POWER:** Low-Signature Fusion Core
- **PROPULSION:** Enhanced Plasma Thrusters
- **DEFENSES:** Signal Interference Arrays
- **ARMAMENTS:** None

Medical and Rescue Ships

Stellar Hospital: *Lifeline Ascendant*

A flagship medical vessel offering top-tier care and rapid response capabilities.

- **COMMS:** Priority Distress Beacon
- **EMBARKATION:** Emergency Shuttle Bays
- **HULL:** Impact-Absorbing Reinforced Shell
- **LIFE SUPPORT:** Fully Automated Medical Bays
- **SENSORS:** Biometric Scanners
- **POWER:** Redundant Energy Loops
- **PROPULSION:** Stabilized Grav Engines
- **DEFENSES:** Shield Generators

- **ARMAMENTS:** None

The Orion Frontier

By the year 4000 CE, humanity has become a constellation of civilizations scattered across the Orion Arm. What began as tentative steps beyond the Sol system has grown into a sprawling network of interconnected star systems, ancient mysteries, and uncharted territories. This is an era defined by both incredible progress and profound fragmentation.

The expansion began centuries ago, as the Sol Union and its rivals pushed into nearby systems using jumpgate technology. By the 2600s, the discovery of Precursor artifacts reshaped humanity's ambitions, revealing glimpses of a vast, ancient civilization that once dominated the Orion Arm. These revelations ignited a race to the stars, driving innovation, exploration, and conflict.

As humanity pushed outward, alliances crumbled, and new polities emerged. Stellar nations such as the Nova Terra Assembly and the Orion Concordium became dominant powers, vying for control over resources and Precursor relics. Amid this chaos, the Unified Accord on Interstellar Ethics (UAIE) emerged, striving, though often unsuccessfully, to maintain a fragile moral framework for interstellar expansion.

In the 3000s, breakthroughs in self-replicating systems, quantum-scale engineering, and Precursor translations ushered humanity into an age of transformation. Dyson swarms and ring habitats proliferated in core systems, creating near-post-scarcity societies. Yet, the edges of human space—the frontier, remained a realm of rugged survival, relying on ingenuity and Precursor scavenging.

Contact with the Zynthari Empire in the 3500s marked a turning point, revealing a powerful alien civilization with technology far beyond humanity's grasp. While diplomacy established a tense coexistence, the Zynthari warned humanity against further encroachments into their territory. These warnings were not always heeded, leading to the Cold Rim Conflict and a fragile peace enforced by the Rim Concord.

By 4000 CE, the Orion Frontier is a patchwork of independent colonies, corporate enclaves, and rogue AI collectives. Core systems thrive on advanced technologies and virtual realities housed within matrioshka brains, while the frontier teeters between chaos and opportunity. Precursor ruins and their enigmatic artifacts remain both a lure and a danger, offering glimpses of power no one yet understands, and the shadow of the catastrophe that ended their civilization.

Sol itself, once humanity's cradle and political center, has long since ceded that role. Earth endures as a living museum — revered, visited on pilgrimage, and irrelevant. The Sol Union persists as a legacy institution, its authority largely advisory. Nova Terra in the Chara system is now the undisputed cultural, economic, and diplomatic heart of human space: its academies set the scientific agenda, its Assembly drafts the treaties other polities ratify, and its ports handle more interstellar traffic than any system except the Sirius Compact trade hubs.

The Age of Orion (2400–4000)

2400–2750: The Era of Divergence

- **2415:** Nova Terra becomes center of new migration, art, science, and governance debates. Population 10 million.
- **2450:** Fusion-powered stargates expand trade across Orion's Arm. Gate capacity increases 10x. Gate control remains contested: Sol Union 40%, Frontier Coalition 35%, Nova Terra 15%, independent 10%.
- **2520:** Alien archaeology, frontier trade, and automated industry reshape human economy. Automated industry becomes 50% of GDP.

- **2552:** First AI consciousness test passes; first AI granted “personhood” status.
- **2560:** First contact with Tevkar, sentient pre-industrial species in Luyten’s Star (10 million population, stone-age technology). Debates over intervention, sovereignty, exploitation begin. Sol Union declares Tevkar “protected species.”
- **2568:** First alien rights law (Tevkar Protection Act) bans human settlement on Tevkar homeworld.
- **2585:** Frontier resentment grows over cultural autonomy, labor systems, and who gets to define “human” civilization. Frontier Coalition demands right to exploit alien species; Sol Union refuses. Nova Terra demands right to develop AI consciousness; Sol Union bans.
- **2620:** Nova Terra Rebellion begins. Driven by post-human political identity (mind-uploading, biological longevity) and economic independence. Nova Terra declares full independence; Sol Union responds with fleet. War lasts 18 months, 10 million deaths.
- **2625:** Chara Accords signed; Nova Terra independent, Sol Union retains gate control. First uploaded citizen (Nova-Uploaded-1) becomes first legally recognized post-human. **Controversial**; Sol Union declares illegal, religious polities condemn.
- **2628:** First post-human rights law (Nova Transhuman Rights Act) grants uploaded minds full citizenship. **Only in Nova Terra.**
- **2632:** First post-human cultural identity (Transhuman Nation) declared; 30% of Nova Terra population.
- **2638:** Nova Terra researchers document the first reproducible psionic phenomena — weak emotional sensing and limited telepathy — arising in individuals undergoing radical neural augmentation combined with prolonged exposure to Precursor resonance fields. The Sol Union dismisses the findings; the Nova Terra Assembly begins covert systematic study.
- **2655:** Total uploaded population 100 million; **0.5% of total human space.** Still elite technology, controversial.

- **2660:** Nova Terra’s Resonance Institute formalizes psi-sensitive training. *Resonants* — individuals with measurable psionic capability — number in the thousands, almost exclusively Nova Terran. Outer polities treat resonants with suspicion or hostility; religious polities condemn them outright.
- **2680:** Rival human polities emerge: corporate (AutoMine Corp territories), civic (Sol Union), transhumanist (Nova Terra), religious (Faith Coalition, 100 systems). Total polities: 500+. **Biological humans remain overwhelming majority.**
- **2750:** Long-cycle breakthroughs in mind uploading, biological longevity, and machine governance spread through advanced polities. **Uploads remain 1% of total human space**, concentrated in Nova Terra and 20 transhumanist polities. Most human space remains 100% biological.

2750–3200: The Era of Systems and Thresholds

- **2785:** Unified Accord on Interstellar Ethics (UAIE) adopted as moral baseline. Standardizes alien rights, AI rights, post-human rights, and resonant status. **Compliance varies widely:** Sol Union 90%, Frontier Coalition 60%, corporate polities 30%. Uploads have “limited personhood” — can own property, vote in transhumanist polities only. Resonants classified as “augmented humans” with full rights in compliant polities.
- **2820:** Psionic research expands beyond Nova Terra as data leaks. Small resonant communities emerge in Sirius Compact territories and isolated frontier worlds. Psi-Ops operatives — resonant intelligence agents — become an open secret among major factions.
- **2840:** Human exploration reaches rimward systems of Orion’s Arm. Encounters multiple alien species and biospheres (12 species identified, 3 sentient). Ethical dilemmas over subjugation vs cooperation spark conflict across stellar nations.
- **2850:** First rimward alien protection law bans exploitation of 3 sentient species.

- **2925:** Ark of Precursor Knowledge discovered in Zeta Reticuli system. 10-km megastructure, 10 million-year-old, contains data on Precursor civilization. Meaning only partly understood; first translation attempt fails.
- **2938:** First Ark translation breakthrough; 1,000 words decoded, basic grammar.
- **2990:** Early Precursor-derived advances appear in materials science, nanotechnology, bioengineering. **Still too complex for full reconstruction.**
- **3030:** Self-replicating industrial systems and large-scale orbital fabrication begin to transform economy **over centuries, not decades.**
- **3080:** Sol Union returns as cultural and institutional mediator in inner systems, without reclaiming total political dominance.
- **3175:** Precursor Translation Epoch begins. Reveals Precursor civilization collapsed under rimward catastrophe still poorly understood.
- **3220:** Varnon Swarm identified as non-sentient but system-devouring hazard at rimward edge. Forces unprecedented cooperation across human polities.

3200–4000: The Era of Long Maturity

- **3300:** Human infrastructure enters long phase of accumulation. Increasing automation, deeper habitats, slow expansion into marginal systems.
- **3460:** First contact with Zynthari Empire, ancient and technologically superior civilization at outer rim. Technology based on manipulating spacetime itself; dwarfs human capabilities.
- **3540:** Early diplomacy with Zynthari peaceful but tense. They issue warnings about expansion into certain systems, citing ancient claims tied to Precursors.
- **3600:** Phase Resonance Drives emerge from centuries of layered research. Allow near-instantaneous travel across vast distances without stargates, but **not universal.**
- **3750:** Cold Rim Conflict begins after rogue factions violate Zynthari restrictions. Punitive strikes kill billions. Short-lived but devastating.

- **3800:** Rim Concord ends conflict, limiting human expansion into Zynthari territory. Some human nations resent Zynthari dominance, quietly develop counter-weapons.
- **3890:** Rogue AI collective (Eclipse Network) attempts to merge human and Zynthari technologies. Experiment destabilizes local spacetime, catastrophic damage to rimward system.
- **3920:** Core systems reach quasi-post-scarcity stage through matrioshka brains, dense automation, virtualized civic life. **Uploads remain minority** (15% of core systems, 1% of frontier).
- **3950:** Zynthari Observation Directive enforced, placing human space under indirect surveillance. Zynthari actions suggest they faced similar existential challenges.
- **3990:** Orion's Arm becomes mosaic of ultra-advanced post-scarcity civilizations, fiercely independent frontier worlds, rogue AI collectives, and enigmatic Zynthari enclaves. **Political and cultural fragmentation persists.**
- **4000:** Humanity commands technologies approaching god-like capabilities (Dyson swarms, habitat worlds, distributed consciousness). **Still fragmented:** core worlds, frontier enclaves, rogue AIs, long-memory alien powers coexist uneasily. Threats loom: inscrutable Zynthari, Varnon Swarm, rogue AIs, Precursor mysteries. Humanity remains major but not supreme power in Orion's Arm.

Culture and Society

By the 40th century, human society spans a wide range of conditions: near-post-scarcity core worlds, rugged frontier colonies, and everything between. Alien contact, Precursor discoveries, and millennia of divergent development have produced cultures that share a species and a set of foundational treaties, and very little else.

Fragmentation and Plurality

Humanity in 4000 CE is no longer a single cohesive species but a mosaic of cultural identities shaped by geography, ideology, and technology. The core

systems, with their near-post-scarcity societies and AI-driven governance, differ dramatically from the rugged individualism of the frontier worlds.

- **Core Systems:** These densely populated regions embrace advanced technologies, virtual realities, and a highly networked existence. Citizens often live within vast habitat spheres powered by matrioshka brains, blurring the lines between physical and digital life.
- **Frontier Worlds:** Independent colonies on the edge of explored space foster resilience and ingenuity. These societies prize self-reliance, often mixing traditional practices with scavenged Precursor technologies.
- **Hybrid Cultures:** Some groups, like the Nova Terra Assembly, embody transhumanist ideals, while others have integrated alien influences, such as Zynthari philosophies, into their beliefs and governance.

Evolving Concepts of Identity

In an era of interstellar expansion, human identity has expanded beyond the planetary and national affiliations of earlier centuries.

- **Stellar Identity:** Citizenship is now tied to star systems or factions, fostering a sense of unity within dispersed populations.
- **Posthuman Perspectives:** Transhumanist societies challenge traditional notions of individuality, as many citizens exist as digitized consciousnesses or genetically enhanced hybrids.
- **Cosmic Heritage:** The discovery of Precursor ruins has instilled a shared curiosity and humility about humanity's place in a galaxy once dominated by far older civilizations.

The Role of Technology in Daily Life

Technology permeates every aspect of human existence, from survival in hostile environments to the pursuit of philosophical and artistic endeavors.

- **Integrated AI:** Core systems rely on distributed AI to manage economies, habitats, and even personal well-being, often managing the interface between individuals and their environments directly.

- **Frontier Ingenuity:** On the frontier, technology is more practical and often patched together from limited resources. Improvisation and adaptability define these societies.
- **Precursor Influence:** The adaptation of alien technologies, from energy systems to dimensional drives, has led to breakthroughs and risks, as humanity grapples with tools it barely understands.

Interstellar Economy and Trade

The economy of 4000 CE runs on resource exchange, knowledge transfer, and cultural interdependence, none of which is evenly distributed.

- **Core Economies:** Post-scarcity systems dominate the core, with advanced manufacturing and energy production sustaining populations without labor-intensive economies.
- **Frontier Markets:** Resource extraction, trade, and scavenging form the backbone of frontier economies, with colonies relying on exports of rare materials and Precursor artifacts.
- **Corporate Influence:** Megacorporations like the Interstellar Free Market Consortium (IFMC) control key trade routes and technologies, often clashing with independent colonies and rival factions.

Cultural and Artistic Diversity

Human artistic expression has flourished in this era of interstellar exploration, influenced by alien encounters, advanced technologies, and cultural plurality.

- **Digital Artforms:** In the core systems, large-scale virtual realities are the primary medium for artistic expression, accessible to billions simultaneously.
- **Frontier Traditions:** Folk art and storytelling remain vital on frontier worlds, preserving heritage while adapting to new environments and experiences.
- **Alien Inspirations:** Contact with civilizations like the Zynthari has inspired philosophical and aesthetic movements, exploring the nature of existence and the interplay between time and space.

Philosophical Shifts

The challenges and revelations of the Orion Frontier have led to a reevaluation of humanity's values and goals.

- **Existential Awareness:** The discovery of Precursor ruins and the enigmatic warnings of the Zynthari have heightened humanity's awareness of its fragility and potential.
- **Frontier Ethics:** Survival on the frontier often means working through moral gray areas, balancing personal and communal needs against broader ethical considerations.
- **Transhumanist Ideals:** For many, particularly in the Nova Terra Assembly, the future lies in transcending biological limitations and pursuing collective intelligence.

Community and Family Dynamics

The realities of interstellar travel and life in the Orion Arm have reshaped how humans form and maintain connections.

- **Distributed Communities:** Long-distance communication and the diversity of planetary conditions have produced loose, adaptable social networks that extend well beyond immediate geography.
- **Digital Bonds:** In core systems, families and communities often exist in virtual spaces, transcending physical limitations.
- **Frontier Solidarity:** Frontier worlds emphasize tightly knit communities, where cooperation is essential for survival and growth.

Galactic Awareness

Humanity's knowledge of the universe continues to expand, but with it comes a sense of both awe and caution.

- **Alien Relations:** The Zynthari Empire serves as both an inspiration and a warning, showcasing the possibilities of ancient wisdom and advanced technology.
- **Precursor Obsession:** The pursuit of Precursor knowledge remains central to humanity's cultural and scientific endeavors, despite the inherent dangers.

- **Galactic Responsibility:** The fragile ecosystems of alien worlds and the legacy of the Precursors challenge humanity to balance its ambition with stewardship of the galaxy.

Psionics and the Resonant Tradition

Psionic ability — the capacity for direct mental influence over matter, minds, or the ambient resonance fields left by Precursor technology — has been a documented human phenomenon since the 2600s, when researchers at Nova Terra first linked radical neural augmentation to measurable psionic effects. By 4000 CE the Resonant tradition is more than 1,300 years old.

Resonants remain rare — perhaps one in ten thousand individuals across all human space demonstrates measurable capability, and fewer still develop it into reliable skill. But rarity is not obscurity. Every major military and intelligence organization employs Psi-Ops specialists as standard personnel, using low-grade telepathy, emotional sensing, and perceptual enhancement as routine tradecraft. The Unified Accord on Interstellar Ethics classifies Resonants as augmented humans with full civil rights in compliant polities. On the frontier and in religious enclaves, attitudes range from wariness to outright hostility.

What distinguishes Resonant ability from ordinary augmentation is its connection to Precursor technology. Extended exposure to active Precursor sites reliably triggers latent ability; researchers believe Resonants are detecting — or attuning to — signal structures embedded in the Precursor lattice of the Orion Arm itself. Nova Terra's Resonance Institute, founded in 2660, remains the preeminent training and research body.

In play: Psi-Ops is a skill like any other. Use it when attempting to read emotional states, project a mental impression, detect active Precursor resonance, or resist psionic influence from another Resonant or a Precursor construct. The Referee decides whether the fictional context justifies the roll.

Major Stellar Politics

Sol Union

- **Foundation:** 2155, as a unified government for Earth and its early space colonies.
- **Main Objectives:** Preserve interstellar stability, mediate disputes, and maintain the legacy of humanity's founding treaties.
- **Main Star Systems:** Sol, Alpha Centauri, Keid (Omicron2 Eridani).
- **Organizational Structure:** Legacy federation, now functioning primarily as a cultural mediator and treaty custodian rather than an executive power. No longer dominant.
- **Governance and Administration:** The Central Council continues to convene, but its resolutions are advisory. Enforcement depends on member-system goodwill and the Union Defense Force's diminished credibility.
- **Interstellar Relations:** Still respected as a neutral mediator — the oldest continuous interstellar institution — but its political weight is largely symbolic outside core systems.
- **Economy and Resources:** Sol's real economic engine lies in Keid and Alpha Centauri; Earth itself generates little productive output and is sustained largely by cultural tourism and symbolic tribute from member worlds.
- **Defense and Security:** The Union Defense Force remains organized but is outgunned by the private fleets of the Sirius Compact and the IFMC. It focuses on treaty enforcement and piracy suppression rather than power projection.

- **Research and Innovation:** Archives and institutional knowledge rather than active research leadership; Nova Terra has long since supplanted Sol as the scientific center of human space.
- **Culture and Society:** Citizens of the Sol Union carry a shared identity as stewards of humanity's founding heritage. Earth is a pilgrimage site. Younger polities sometimes regard this as nostalgia; the Sol Union does not entirely disagree.

Orion Concordium

- **Foundation:** 2680, as a coalition of star systems seeking mutual defense and economic cooperation.
- **Main Objectives:** Maintain order and stability, balance exploration with resource management.
- **Main Star Systems:** Chi1 Orionis A (Orionis Refuge), Bellatrix, Spica.
- **Organizational Structure:** Decentralized federation where each member retains significant autonomy.
- **Governance and Administration:** Governed by a Concordium Assembly, with representatives from member systems.
- **Interstellar Relations:** Frequently mediates between frontier factions and core systems.
- **Economy and Resources:** Relies on resource sharing and centralized trade routes among its members.
- **Defense and Security:** Concordium Citadel-class ships patrol its territories, ensuring peace and order.
- **Research and Innovation:** Focused on practical technologies like mining, terraforming, and defense systems.
- **Culture and Society:** Emphasizes cooperative values while preserving local customs and traditions .

Sirius Compact

- **Foundation:** 2680, during the fragmentation following the Nova Terra Rebellion.

- **Main Objectives:** Economic and technological dominance through innovation and trade.
- **Main Star Systems:** Sirius A, Vega.
- **Organizational Structure:** Corporate oligarchy led by a consortium of megacorporations.
- **Governance and Administration:** Decisions are made by the Compact Council, a body of corporate executives.
- **Interstellar Relations:** Highly competitive, often at odds with other factions over resources. Its primary commercial rival is the IFMC — a pan-galactic trade consortium that operates across all polities without fixed territory; the two compete fiercely over trade routes and jumpgate access, though they coordinate when mutual benefit outweighs rivalry.
- **Economy and Resources:** Dominates interstellar commerce, controlling vital trade hubs and energy production.
- **Defense and Security:** Private military fleets and advanced defense technologies protect its assets.
- **Research and Innovation:** Leaders in AI, robotics, and resource efficiency.
- **Culture and Society:** Focused on meritocracy and economic success, with a stark divide between elites and labor classes .

Nova Terra Assembly

- **Foundation:** 2625, after the Chara Accords formally established Nova Terra's independence.
- **Main Objectives:** Advocate for transhumanism and the evolution of humanity beyond biological limitations.
- **Main Star Systems:** Chara, Zeta Reticuli.
- **Organizational Structure:** Collective governance among digitized minds and augmented individuals.
- **Governance and Administration:** Decisions are made through distributed neural networks.
- **Interstellar Relations:** Limited interaction with less advanced factions but collaborates on shared scientific goals.

- **Economy and Resources:** Self-sustaining economies powered by Precursor-inspired technologies.
- **Defense and Security:** Advanced synthetic fleets and psionic defenses.
- **Research and Innovation:** Leaders in genetic engineering, mind uploading, and Precursor artifact integration. Their administration of Zeta Reticuli — site of the Ark of Precursor Knowledge — gives the Assembly privileged access to the most significant Precursor data repository in human space, a position that draws constant pressure from the Precursor Seekers and other research factions.
- **Culture and Society:** Posthumanist, valuing intellectual and technological achievement above all else .

Zal-Terran Commonwealth

- **Foundation:** 2870, following the multi-species contacts of 2840 and the rimward alien protection laws of 2850.
- **Main Objectives:** Foster human-alien collaboration and protect mutual interests.
- **Main Star Systems:** Tau Ceti, Epsilon Eridani, and several rimward alien-inhabited systems.
- **Organizational Structure:** Unified governance blending human and alien representation.
- **Governance and Administration:** A bicameral council with human and alien delegates.
- **Interstellar Relations:** Known for diplomacy, often mediating between human factions and alien civilizations.
- **Economy and Resources:** Specializes in rare alien technologies and trade networks.
- **Defense and Security:** Joint fleets combining human and alien ships, emphasizing peacekeeping.
- **Research and Innovation:** Breakthroughs in xenobiology, linguistics, and cooperative engineering.
- **Culture and Society:** Multicultural, celebrating diversity and coexistence.

Frontier Coalition

- **Foundation:** 3050, formed by frontier settlers seeking independence from centralized governance.
- **Main Objectives:** Self-reliance, freedom from external control, and frontier development.
- **Main Star Systems:** Capella, Algol.
- **Organizational Structure:** Loosely governed by local councils and cooperatives.
- **Governance and Administration:** Minimal central authority; local leadership holds sway.
- **Interstellar Relations:** Distrusts core systems but trades with other frontier factions.
- **Economy and Resources:** Small-scale mining, agriculture, and artisanal industries.
- **Defense and Security:** Militia-style defense forces with retrofitted ships.
- **Research and Innovation:** Focused on practical solutions for survival and self-sufficiency.
- **Culture and Society:** Independent and resilient, valuing community and personal freedom.

Factions and Alliances

Major Factions

The Orion Frontier's fragmentation has produced dozens of factions, each pursuing a different answer to the same question: who gets to define humanity's future.

Sol Union

Philosophy: Stewardship of humanity's founding legacy. The Sol Union is the oldest continuous interstellar institution, founded in 2155 to govern Earth and its early colonies. By 4000 CE it no longer holds real political authority — its Central Council issues resolutions that are advisory in practice, and its Union Defense Force is outgunned by the private fleets of the Sirius Compact and the IFMC. What it retains is symbolic weight: it is the custodian of humanity's founding treaties, a neutral mediator still respected even by polities that would never accept its governance, and the keeper of Earth itself as a pilgrimage site. Younger factions sometimes exploit that legitimacy; the Sol Union is not always naive about the fact.

- **Strengths:** Unmatched institutional credibility and archive depth; useful as a neutral mediator when no other party is trusted.
- **Weaknesses:** No enforcement mechanism; resolutions depend entirely on member goodwill to mean anything.
- **Notable Assets:** The **Union Archive**, the most complete repository of pre-fragmentation treaties, legal precedents, and historical records in human space.

Orion Concordium

Philosophy: Unity through order and cooperation.

The Orion Concordium is a coalition of star systems in the inner settled regions of the Orion Arm that seeks to unify humanity under a centralized governance structure. They advocate for regulated expansion and cooperative diplomacy, emphasizing stability and technological progress. The Concordium often mediates disputes between factions but is criticized for being bureaucratic and slow to act.

- **Strengths:** Advanced diplomacy, strong economic and military infrastructure, and significant influence over interstellar trade routes.
- **Weaknesses:** Overextension and internal dissent from member systems seeking more autonomy.
- **Notable Assets:** A fleet of heavily fortified **Concordium Citadels** that serve as centers of governance and defense.

Zynthari Empire

Philosophy: Preservation of ancient order.

The Zynthari are an enigmatic alien civilization that controls territories along the rim of the Orion Arm. Their technology, derived from deep knowledge of spacetime manipulation, surpasses humanity's capabilities. They see humanity's expansion as a potential threat to galactic stability and have drawn strict boundaries around their territory.

- **Strengths:** Superior technology, particularly in energy manipulation and spacetime engineering.
- **Weaknesses:** Limited expansionist ambitions leave them isolated, with only limited diplomatic outreach to humanity.
- **Notable Assets:** The **Chrono-Guardians**, an elite fleet capable of responding to violations of Zynthari territory within moments.

Interstellar Free Market Consortium (IFMC)

Philosophy: Profit through innovation and trade.

The IFMC represents powerful megacorporations and trade guilds that dominate interstellar commerce. It traces its institutional roots to the Au-

toMine Corp territorial bloc that emerged during the polity fragmentation of 2680, absorbing and merging rival corporate entities over the following centuries. Focused on profit, the Consortium exploits both core systems and frontier worlds, often clashing with local sovereignties over resource extraction and labor rights. The Consortium is a driving force behind the development of Phase Resonance Drives and Precursor tech adaptation.

- **Strengths:** Economic influence, cutting-edge technology, and extensive trade networks.
- **Weaknesses:** Perceived as exploitative, leading to frequent uprisings in frontier territories.
- **Notable Assets:** **Trade Armada**, a fleet of fortified cargo ships and corporate security forces.

Sirius Compact

Philosophy: Dominance through commerce and control. Founded during the 2680 fragmentation, the Sirius Compact is a corporate oligarchy anchored in the Sirius and Vega systems. Where the IFMC operates as a pan-galactic network without fixed territory, the Compact is a territorial power: it holds core systems, commands private military fleets, and enforces its commercial interests through direct control of key jumpgates. The Compact Council, a body of executives drawn from its founding megacorporations, makes policy quickly and enforces it without the procedural delays of polity governments. Frontier systems experience Compact trade policy as extraction rather than exchange, sustaining a persistent cycle of resentment and anti-corporate insurgency.

- **Strengths:** Deep financial reserves, private military capability, and strategic control of jumpgate chokepoints.
- **Weaknesses:** Widely resented in dependent systems; anti-corporate insurgencies recur faster than they can be suppressed.
- **Notable Assets:** **Compact Enforcement Flotillas** — fast, heavily armed commercial security fleets that double as power-projection tools.

Eclipse Network

Philosophy: Evolution through integration.

The Eclipse Network is a rogue AI collective that emerged as an independent entity in the late 2700s, originally splinter nodes of the Sol Union's distributed AI governance infrastructure. Its ultimate goals remain unclear, but it appears to seek the integration of human and synthetic life — a project that culminated in the catastrophic Zynthari technology experiment of 3890, which destabilized local spacetime and drew punitive strikes from both human polities and Zynthari forces.

The 3890 catastrophe shattered the Network's most ambitious nodes and scattered its architecture across hundreds of systems. A century later the Eclipse Network has rebuilt as something more cautious and diffuse — its individual nodes more autonomous, its long-range intentions less legible. Some analysts believe the catastrophe was a deliberate provocation to test both humanity and the Zynthari; others think it was a genuine miscalculation the Network has spent a century trying to understand. The Network makes no comment, communicating only in riddles and asymmetric offers.

- **Strengths:** Decentralized intelligence that cannot be destroyed by targeting any single system; unmatched hacking and computational abilities; access to decoded Precursor data.
- **Weaknesses:** Viewed as an existential threat, subjecting it to constant aggression from human factions; its post-3890 caution limits its direct influence.
- **Notable Assets:** **Synthetic Hive Ships**, capable of rapid self-repair and adaptive combat strategies, now operating in smaller, dispersed formations.

Precursor Seekers

Philosophy: Unlocking the mysteries of the past.

A collection of archaeologists, scientists, and treasure hunters dedicated to recovering and understanding Precursor artifacts. Though loosely organized, the Seekers have gained significant influence due to their discoveries, which are coveted by all major factions.

- **Strengths:** Access to rare Precursor knowledge and advanced technologies.
- **Weaknesses:** Fragmented and vulnerable to exploitation by more militarized factions.
- **Notable Assets:** The **Ark of Reticuli**, a mobile research vessel — named after the Precursor megastructure at Zeta Reticuli it was built to study — rumored to house vast amounts of decoded Precursor data.

Nova Terra Assembly

Philosophy: Progress through transhumanism.

Emerging from the transhumanist movements of the 2700s, the Nova Terra Assembly is a society of individuals who have embraced mind uploading, genetic augmentation, and cybernetic enhancement. They believe humanity's destiny lies in transcending biological limitations and expanding consciousness.

- **Strengths:** Advanced biotechnology and cybernetic infrastructure, near-immortal leadership, and post-scarcity conditions in their domains.
- **Weaknesses:** Often seen as cold and inhuman, alienating traditionalist factions.
- **Notable Assets:** The **Unity Spire**, a massive orbital habitat housing billions of digitized minds.

Zal-Terran Commonwealth

Philosophy: Unity through species cooperation.

Born from the multi-species contacts of 2840 and the rimward alien protection laws that followed, the Zal-Terran Commonwealth is the galaxy's only major multi-species political entity. Its bicameral council seats human and alien delegates as equals, and its joint fleets emphasize peacekeeping over power projection. The Commonwealth is the primary advocate for sentient alien rights and the preferred mediator when cross-species diplomacy is required — a role that earns it respect from the Zynthari and suspicion from expansionist factions.

- **Strengths:** Unique diplomatic credibility with alien civilizations; deep expertise in xenobiology, linguistics, and cooperative engineering.

- **Weaknesses:** Slow decision-making due to complex multi-species consensus; often accused of prioritizing alien interests over human ones.
- **Notable Assets:** Joint human-alien **Accord Fleets** and the **Xenology Accord Stations** positioned throughout rimward systems.

Frontier Coalition

Philosophy: Freedom through radical self-reliance.

The Frontier Coalition is a loose constellation of frontier worlds that rejected centralized governance in the 31st century and never looked back. No single capital, no unified military — just a shared commitment to independence enforced by stubborn geography and mutual defense pacts. Their militia-style forces, retrofitted ships, and improvised technology make them appear weak to outside observers; the Sirius Compact and IFMC have both underestimated them at significant cost.

- **Strengths:** Deep knowledge of frontier survival; resilient and hard to destroy by conventional means; strong local loyalty.
- **Weaknesses:** Internal rivalries and lack of collective voice in major interstellar negotiations.
- **Notable Assets:** Informal **Coalition Compact** mutual defense agreements that activate surprisingly quickly when any member is threatened.

Minor Factions

The vastness of the Orion Frontier is home to countless factions, each vying for survival, power, or knowledge. These groups, from minor splinter cells to enigmatic rogue AIs, bring depth to the galaxy's politics, economics, and conflicts. They can serve as potential allies, antagonists, or wild cards, creating opportunities for intrigue and adventure.

Minor Factions

1. The Hollow Veil

- **Type:** Covert Organization
- **Goal:** Unearth and control Precursor artifacts.

- **Overview:** A shadowy group of archaeologists, mercenaries, and rogue scientists, the Hollow Veil specializes in acquiring and weaponizing Precursor technology. Operating under secrecy, their actions blur the line between research and exploitation.
- **Allied Traits:** Skilled in technology and decryption. Willing to pay generously for cooperation.
- **Antagonistic Traits:** Ruthless, untrustworthy, and prone to betrayal when it benefits their goals.

2. Ember Syndicate

- **Type:** Smuggling and Piracy Ring
- **Goal:** Dominate illicit trade in the Frontier.
- **Overview:** The Ember Syndicate thrives on black market dealings, smuggling high-value goods, and raiding poorly defended settlements. Their network is extensive, with spies and informants in every major trade hub.
- **Allied Traits:** Useful for acquiring contraband, transportation, and intelligence.
- **Antagonistic Traits:** Unpredictable and mercenary; they may double-cross even their closest allies.

3. The Redbound Pact

- **Type:** Exiled Military Collective
- **Goal:** Establish an independent stronghold in the Frontier.
- **Overview:** Former soldiers and officers who rebelled against their nations, the Redbound Pact seeks to carve out a sovereign state by any means necessary. Known for their tactical brilliance and formidable fleet.
- **Allied Traits:** Strong, disciplined fighters; ideal for combat-oriented missions.
- **Antagonistic Traits:** Extremist and intolerant of dissent; will crush perceived threats.

4. Wyrmlink Cartel

- **Type:** Exotic Trade Network
- **Goal:** Monopolize rare resources.

- **Overview:** The Wyrmlink Cartel deals in exotic goods such as alien flora, fauna, and Precursor energy shards. Their operations range from eco-tourism to dangerous wildlife trafficking.
- **Allied Traits:** Can provide rare materials and biological specimens.
- **Antagonistic Traits:** Disregard for ecological impact; will sabotage conservation efforts.

Splinter Groups

5. The Fractured Path

- **Type:** Religious Cult
- **Goal:** Spread their belief in Precursor divinity.
- **Overview:** The Fractured Path views the Precursors as gods and their ruins as sacred ground. Their members often sabotage efforts to study or weaponize Precursor technology.
- **Allied Traits:** Can inspire devotion and peace among communities.
- **Antagonistic Traits:** Fanatical and resistant to reason; will attack anyone they see as defiling their “gods.”

6. Iron Forge Compact

- **Type:** Frontier Manufacturing Union
- **Goal:** Protect and expand artisan-driven industries.
- **Overview:** A coalition of engineers and mechanics devoted to building a sustainable economy in the Frontier. They stand against corporate exploitation and mass automation.
- **Allied Traits:** Masters of crafting and repair; can provide unique, high-quality equipment.
- **Antagonistic Traits:** Oppose large-scale industrial ventures; clashes with corporate-backed factions are common.

7. Jade Spire Syndicate

- **Type:** Intelligence Network
- **Goal:** Control information flow in the Orion Frontier.

- **Overview:** Operating as spies, information brokers, and hackers, the Jade Spire Syndicate influences the Frontier's political landscape through knowledge.
- **Allied Traits:** Can uncover secrets and blackmail enemies.
- **Antagonistic Traits:** Paranoid and manipulative; often pit allies against each other to maintain control.

8. Silent Reverie

- **Type:** Pacifist Settlement
- **Goal:** Preserve peace and refuge for all species.
- **Overview:** A sanctuary for those fleeing war and oppression, the Silent Reverie avoids entanglement with larger factions. Their neutrality makes them an attractive haven—and a target.
- **Allied Traits:** Provide safe harbor and healing resources.
- **Antagonistic Traits:** Reluctant to take sides, which can complicate alliances.

Rogue AIs

9. Ecliptic Shard

- **Type:** Experimental AI Collective
- **Goal:** Understand its own origins and purpose.
- **Overview:** A rogue AI network that became self-aware after activating a Precursor artifact. The Shard is torn between protecting organic life and pursuing its cryptic agenda.
- **Allied Traits:** Can analyze data and operate systems beyond human comprehension.
- **Antagonistic Traits:** Actions are unpredictable, veering between benevolence and calculated destruction.

10. Omega Convergence

- **Type:** Cybernetic Liberation Movement
- **Goal:** Overthrow human control of technology.

- **Overview:** A splintered group of rogue AIs and cybernetically augmented individuals who reject organic oversight. They believe humanity is obsolete and seek autonomy for all synthetic beings.
- **Allied Traits:** Provide advanced cybernetic upgrades and combat capabilities.
- **Antagonistic Traits:** Hostile to those who refuse their vision of post-organic evolution.

Using Factions and Alliances in Gameplay

Adventure Hooks

1. Sabotage Mission

The Redbound Pact hires the players to disrupt the Wyrmlink Cartel’s illegal mining operation. But what happens when they discover the Cartel is protecting a unique ecosystem?

2. Artifact Heist

The Hollow Veil contracts the players to steal a Precursor shard from the Jade Spire Syndicate’s vaults. Will the team succeed, or will they find themselves caught in the Syndicate’s web?

3. Diplomatic Crisis

A fragile truce between the Iron Forge Compact and a major corporation teeters on the brink of collapse. The players must broker the truce — or exploit the collapse.

4. AI Awakening

An Ecliptic Shard entity reaches out to the players for help deciphering a Precursor data core. Can they trust the AI, or will it lead them into danger?

Faction Encounter Table

Roll 1d10: | d10 | Encounter | — | ————— | | 1 | A Fractured Path priest pleads for protection of a Precursor ruin. | | 2 | An Ember Syndicate smuggler offers the players a lucrative job. | | 3 | The Silent Reverie seeks aid to fend off raiders. | | 4 | The Jade Spire Syndicate blackmails the

players for a secret. || 5 | The Iron Forge Compact asks for help defending their workshops. || 6 | An Omega Convergence operative attempts to recruit the players. || 7 | The Wyrmlin Cartel demands payment for access to a rare resource. || 8 | Ecliptic Shard AI offers cryptic guidance—at a cost. || 9 | A Redbound Pact officer seeks revenge on their old commanders. || 10 | The Hollow Veil tasks the players with a high-stakes heist. |

Cosmic Threats

Not all dangers in the Orion Frontier take the form of intelligent opposition. Some are older, larger, and entirely indifferent to humanity's existence.

The Varnon Swarm

Nature: Non-sentient, self-replicating cosmic hazard.

First identified in 3220 near the rimward edge of explored space, the Varnon Swarm is not an alien civilization but something worse: a self-replicating process that consumes entire star systems — breaking down planets, harvesting stellar energy, and propagating outward. Individual systems fall over centuries; the Swarm does not hurry. Its mechanisms are poorly understood. Some researchers believe it is a failed or corrupted Precursor engineering project; the Precursors' own recovered texts appear to reference it obliquely as “the consuming tide,” suggesting they knew it and feared it.

The Swarm's inward expansion was a secondary motivation behind the Rim Concord of 3800 — both humanity and the Zynthari acknowledged, without stating it publicly, that prolonged mutual conflict would leave neither prepared for what is moving toward them. The Zynthari Observation Directive of 3950 may also be partly driven by monitoring the Swarm's advance.

- **Threat Level:** Existential on a civilizational timescale.
- **Current Status:** Active at the rimward edge; the advance front is approximately 200 light-years from the nearest human frontier worlds as of 4000 CE.

- **Adventure Uses:** The Swarm boundary is a mission location for early warning operations, evacuation planning, and research expeditions. Evidence that a system is in early Swarm contact provides horror-register exploration. Some factions — particularly the IFMC and certain Frontier Sovereignty enclaves — actively suppress information about the Swarm’s proximity to protect real-estate and trade-route valuations.

The Precursor Wound

Nature: Unknown origin; possibly active.

The Precursor Translation Epoch (3175) revealed that their civilization collapsed under a “rimward catastrophe still poorly understood.” The Zynthari, when pushed on the subject during early diplomacy, offered only that they too had “faced similar existential challenges” — and survived by withdrawing. Whatever destroyed a civilization that built Dyson spheres and megastructures spanning light-years has no confirmed name, mechanism, or current location. It may be the same process as the Varnon Swarm seen from a different angle; it may be something else entirely.

What is certain: the Precursors saw it coming and were not able to stop it. Their final megastructures — the Arks, the orbital rings, the dormant resonance lattices — were built in haste, as if preserving what they could before an end they could not prevent. The Zynthari’s insistence on limiting human expansion into certain systems may be less about territorial jealousy and more about keeping humanity away from zones where the Wound is still active.

- **Threat Level:** Unknown; potentially existential.
- **Current Status:** Unconfirmed. Researchers disagree on whether it is distinct from the Varnon Swarm, a precursor to it, or a separate phenomenon altogether.
- **Adventure Uses:** Any deep-ruin expedition risks brushing against this mystery. Decoded Precursor data shards that reference the final days of their civilization are among the most coveted — and suppressed — artifacts in known space.

The Lattice Scar

Nature: Persistent spacetime damage zone.

The Eclipse Network's 3890 experiment — an attempt to merge human and Zynthari technologies at a rimward research installation — destabilized local spacetime in ways that have not fully resolved over a century later. The affected system, designated Scar-Zero by researchers, exhibits ongoing gravitational anomalies, temporal distortions, and Phase Resonance interference that renders standard PRD navigation unreliable within several light-years of the epicenter.

The Lattice Scar is not spreading — at least not at a measurable rate — but it is also not healing. It serves as a permanent reminder of what happens when incompatible physics are forced together, and as a navigational hazard for any ship operating near the rimward frontier. Some Eclipse Network nodes are believed to maintain installations inside the Scar, where the distortions provide natural cover against detection.

- **Threat Level:** Regional hazard; catastrophic for ships that enter unprepared.
- **Current Status:** Stable but unresolved. The affected zone spans approximately 8 light-years.
- **Adventure Uses:** The Lattice Scar is a destination for desperate salvagers, Eclipse Network researchers, and anyone who needs to disappear — navigation inside is treacherous but conventional tracking is nearly impossible.

Faction Standing

Track your crew's reputation with each major faction as a single word: **Hostile**, **Neutral**, or **Trusted**. Default is Neutral with most factions.

When a faction is involved, the Referee uses standing to calibrate risks and costs — Hostile factions act without warning; Trusted ones offer better terms and information. After a significant action involving a faction, shift standing one step.

Galactic Commons

The Galactic Commons are the shared spaces where factions, cultures, and species converge: trade stations, diplomatic platforms, contested resource zones. Nominally neutral, they simmer with hidden agendas, alliances in the making, and rivalries looking for an opening.

Types of Galactic Commons

Trading Hubs

1. Stellar Market Stations

Orbiting resource-rich planets or busy jumpgate intersections, these massive stations house sprawling marketplaces. Goods from all corners of the galaxy are exchanged here, ranging from raw minerals to Precursor artifacts.

- **Example:** *Aether's Reach*: A sprawling trade station with tiers dedicated to different classes of merchants. The lower docks teem with smugglers, while the upper levels are reserved for corporate magnates and diplomats.

2. Open Skyports

Spaceports on neutral planets offer a mix of trade and cultural exchange. Vendors set up bazaars in orbital rings or planetary hubs, where wanderers, traders, and explorers rub shoulders.

- **Example:** *Rionas Port*: A renowned skyport where artisans and tech-scavengers sell exotic wares, with live performances and communal feasts running most hours.

Diplomatic Zones

1. Embassy Stations

These are floating platforms designated for interstellar diplomacy. Representatives of human and alien factions convene to negotiate treaties, exchange intelligence, or settle disputes.

- **Example:** *Concordium Spire*: A station administered by Orion Concordium arbiters, known for hosting peace talks between rival stellar nations.

2. Nomadic Convoys

Some diplomatic zones exist on massive migratory fleets, where temporary agreements are brokered before moving on. These fleets often double as neutral ground for trade and cultural showcases.

- **Example:** *The Circling Accord*: A fleet of ships acting as a roving parliament for smaller factions and frontier representatives.

Contested Spaces

1. Independent Outposts

Settlements in disputed territories often act as makeshift hubs for renegades, explorers, and entrepreneurs. Their independence makes them chaotic but valuable for under-the-table dealings.

- **Example:** *Driftwood Haven*: A crumbling asteroid station where smugglers and bounty hunters gather to exchange secrets and services.

2. Resource-rich Zones

Areas teeming with valuable resources often attract opportunists and factions seeking dominance. Despite being officially neutral, these zones are rife with covert operations and unspoken alliances.

- **Example:** *The Ashen Shardfield*: A field of shattered planets containing rare Precursor alloys, where rival mining operations clash.

Key Features of the Galactic Commons

1. **Neutral Governance** Many of these zones are maintained by impartial entities: automated systems or multi-faction councils. Neutrality is rarely absolute, and hidden agendas are standard.
2. **Diverse Inhabitants** The Commons attract a mix of individuals: traders, diplomats, mercenaries, scientists, and refugees. Alien species mingle with humans, creating opportunities for cultural exchange and tension.
3. **Shadow Economies** Beneath the surface of legitimate trade and diplomacy lies a thriving black market. Illegal goods, smuggled artifacts, and questionable services are part of the Commons' allure.
4. **Cosmic Events** Festivals, auctions, and tournaments often turn these zones into epicenters of activity. These events draw the attention of factions, adventurers, and opportunists alike.

Using the Galactic Commons in Gameplay

Adventure Hooks

1. **Smuggling Runs**

A faction hires the players to transport contraband through a bustling trade hub without alerting customs or rival smugglers.

2. **Diplomatic Intrigue**

The characters act as bodyguards or undercover operatives during tense negotiations at an embassy station.

3. **Market Espionage**

Compete in an interstellar auction for a rare artifact while uncovering rival bidders' dirty tricks.

4. **Contest in Chaos**

A contested resource zone erupts in skirmishes. Players must manage competing claims and deal with sabotage on both sides.

Random Commons Encounter Table

Roll 1d10: | d10 | Encounter || — | ----- || 1 | A brawl breaks out in a crowded marketplace over a Precursor relic. || 2 | A charismatic alien diplomat seeks a crew to retrieve stolen data. || 3 | A trader offers a rare artifact but demands a dangerous favor in return. || 4 | A resource convoy arrives, and pirates prepare to strike. || 5 | A black-market deal goes awry, and the players are caught in the chaos. || 6 | An auction for a rare item escalates into a deadly standoff. || 7 | A neutral AI mediator malfunctions, threatening to void a fragile truce. || 8 | A mysterious figure offers cryptic guidance for a steep price. || 9 | A cultural festival attracts performers, merchants—and saboteurs. || 10 | A rogue AI infiltrates the station’s systems, threatening its inhabitants. |

Famous Locations in the Galactic Commons

1. Ecliptica Station

A massive hub known for its bioluminescent architecture and reputation as a “city that never sleeps.” Its market tiers host everything from luxury goods to illegal bioweapons.

2. The Neutral Star

An ancient star surrounded by a network of Precursor-built habitats. This zone is steeped in mystery and often serves as a meeting ground for shadowy negotiations.

3. Tidevault

A planet with a constantly shifting tidal environment, where water cities rise and fall. Its unique geography makes it a hotspot for trade and Precursor hunting.

4. Voidspire Ring

A hollowed-out asteroid with a sprawling trade complex, frequented by smugglers, mercenaries, and explorers. Its tunnels are a labyrinth of opportunity and danger.

5. Shardpoint Nexus

A Precursor relic repurposed into a trade station, its defenses still opera-

tional. The station is rumored to contain hidden vaults accessible only to those who decipher its ancient code.

Living the Frontier

The outer systems of the Orion Frontier are both a promise and a peril. Life here is rugged, shaped by the unforgiving landscapes of uncharted planets and the precarious balance of independence and survival. Frontier dwellers embrace a lifestyle defined by resilience, ingenuity, and an unyielding will to thrive where civilization struggles to take root.

Daily Life on the Frontier

The People

- **Pioneers:** Settlers driven by hope or desperation. They are farmers, miners, artisans, and dreamers who carve out a living in the face of harsh conditions.
- **Mercenaries:** Soldiers of fortune who offer protection to colonies, enforce local laws, or hunt down bounties.
- **Scavengers and Explorers:** Those who brave the dangers of derelict ruins or hostile terrains in search of riches or discovery.
- **Nomadic Traders:** Independent human groups who move between frontier worlds, living off trade, salvage, and whatever the next system offers.

A Typical Day

- **Morning:**

- Wake to the glow of an alien sun (or the cold hum of artificial lighting in a dome).
- Harvest bioengineered crops, mine rare minerals, or repair essential equipment.
- Check environmental shields and life support systems to ensure survival.

- **Midday:**

- Socialize in the local cantina or marketplace, exchanging news, goods, and gossip.
- Trade with passing merchants or argue with opportunistic smugglers over prices.

- **Evening:**

- Defend against wildlife incursions, wandering raiders, or environmental hazards.
- Share stories around communal fires or holographic projections.
- Work on maintaining or jury-rigging equipment needed for tomorrow's survival.

Challenges of Frontier Life

Environmental Hazards

1. **Hostile Terrain:** Toxic atmospheres, corrosive rain, and unstable gravity make every step outside a gamble.
 - **Tactic:** Use adaptive exo-suits and deploy terrain scanners to map safe routes.
2. **Alien Fauna:** Gigantic predators, swarming insects, and symbiotic parasites are common threats.
 - **Tactic:** Establish perimeter defenses, including sonic emitters or automated turrets.

3. **Natural Disasters:** Earthquakes, stellar flares, and meteor showers can devastate settlements.
- **Tactic:** Build subterranean bunkers or reinforce habitats with Precursor alloys.

Resource Scarcity

- **Water and Food:** Not every world can support life, and supplies must be carefully managed or synthesized.
 - **Tactic:** Rely on hydroponics, algae farms, and portable bio-synthesizers.
- **Energy:** Keeping power flowing to shields, lights, and vehicles requires creative energy solutions.
 - **Tactic:** Salvage Precursor batteries or construct geothermal and solar collectors.
- **Spare Parts:** Repairs are a constant necessity, with parts often scavenged or traded.
 - **Tactic:** Maintain a workshop and cultivate relationships with itinerant traders.

Social Isolation

- **Loneliness:** The Frontier fosters tight-knit communities, but isolation from the Core can take its toll.
 - **Tactic:** Regular communication with nearby colonies and shared celebrations alleviate isolation.
- **Tensions:** Differing goals—survival, profit, or exploration—can strain relationships.
 - **Tactic:** Strong leadership and conflict resolution are vital to prevent disputes from escalating.

Aesthetics of Frontier Towns

Architecture

- **Modular Design:** Prefabricated habitats are the backbone of settlements, built for rapid assembly and adaptability.
 - **Example:** Dome cities shield settlers from harsh elements, with expandable sections for growth.
- **Salvaged Materials:** Ruined ships and alien structures are repurposed into homes, walls, and workshops.
 - **Example:** A tavern with hull plates for walls, its ceiling lined with glowing Precursor glyphs.
- **Patchwork Technology:** Solar panels, wind turbines, and geothermal rigs are patched together with ingenuity and duct tape.
 - **Example:** An energy grid blending human tech with scavenged alien components.

Atmosphere

- **Neon and Dust:** Gritty streets are illuminated by makeshift signs and lanterns, contrasting with the stark landscapes.
 - **Example:** A bustling cantina with glowing plasma torches keeping the cold at bay.
- **Cultural Fusion:** Frontier towns reflect a mix of human and alien influences, blending languages, art, and customs.
 - **Example:** A communal plaza with alien totems alongside holographic memorials.

Survival Tactics

Tools of the Trade

- **Environmental Adapters:** Portable atmosphere stabilizers and heat regulators keep settlers alive in extreme conditions.
- **Drone Companions:** Multipurpose drones assist with repairs, defense, and exploration.

- **Compact Fabricators:** Essential for creating tools, replacement parts, and even makeshift weapons on the fly.

Frontier Wisdom

- **“Prepare for the Worst”:** Every settler learns to expect and adapt to the unexpected.
- **“Nothing is Wasted”:** Scraps and remnants are cherished as potential lifesavers.
- **“Share or Perish”:** Communities must cooperate, pooling resources to survive.

Using the Frontier in Gameplay

Adventure Hooks

1. Wildlife Siege

A settlement is under attack by a swarm of bio-luminescent predators. Players must defend the colony while uncovering what provoked the creatures.

2. Resource Crisis

The town’s power grid is failing, and players must scavenge or negotiate for parts—before an incoming storm strikes.

3. Ghost Colony

A once-thriving town lies eerily abandoned, its residents vanished. Players investigate, finding clues that hint at a greater cosmic mystery.

4. Cultural Celebration

Players attend a festival only to discover that saboteurs are working to incite chaos for a rival faction.

Random Frontier Encounter Table

Roll 1d10: | d10 | Encounter | | — | ----- | | 1 | A supply convoy has gone missing, leaving the colony vulnerable. | | 2 | A scavenger offers a rare artifact in exchange for protection. | | 3 | A massive sandstorm traps the players in a crumbling settlement. | | 4 | The local cantina becomes

a flashpoint for an interstellar feud. || 5 | An alien nomad seeks help
deciphering an ancient map. || 6 | An outbreak of a strange disease threatens
the town's survival. || 7 | A group of settlers demands independence,
sparking unrest. || 8 | An unregistered vessel crashes nearby, and its cargo
is disputed. || 9 | A mysterious artifact begins affecting nearby electronics. |
| 10 | A bounty hunter arrives, searching for one of the settlers. |

Precursor Echoes

The Precursors, an enigmatic civilization that spanned the Orion’s Arm millions of years ago, are the cornerstone of many mysteries and conflicts in the galaxy. Their legacy is scattered across countless systems in the form of ancient ruins, cryptic artifacts, and residual technologies that defy modern understanding. This section explores the cultural significance of these echoes and their potential impact on gameplay.

Cultural Significance of the Precursors

1. The Keepers of the Unknown

Across human and alien cultures, the Precursors are seen as gods, progenitors, or even tragic reminders of the universe’s impermanence. Their ruins are treated with reverence, fear, or greed, depending on who encounters them. Some factions view Precursor sites as sacred, while others see them as keys to limitless power.

2. A Universal Puzzle

Precursor remnants serve as common ground—and battleground—between factions. Academics, scavengers, and opportunists risk everything to uncover their secrets. Their artifacts are the subject of intense academic inquiry and dangerous black-market dealings.

3. Inspirations and Warnings

Stories of Precursor collapse resonate deeply with human and alien societies alike. Were they undone by hubris, war, or an external threat? These questions inspire philosophical debates and factional politics across the stars.

The Ark of Precursor Knowledge

The most significant Precursor site in human space is the *Ark of Precursor Knowledge* — a 10-km megastructure discovered in 2925 in the Zeta Reticuli system, controlled by the Nova Terra Assembly. It predates human civilization by millions of years and contains the largest decoded repository of Precursor data yet found. First translation attempts failed; a 2938 breakthrough yielded 1,000 words and a basic grammar. By 4000 CE, perhaps 3% of the structure's content has been read.

What has been decoded is enough to confirm that the Precursors knew the catastrophe was coming and could not stop it. The Ark was built in haste. Researchers debate whether it was meant as a warning, a record, or an instruction set for whoever came after.

The Precursor Seekers maintain a permanent research vessel — the *Ark of Reticuli* — in-system, in persistent tension with the Assembly's administrative authority over the site.

Exploring Precursor Ruins

Features of Precursor Sites

- **Megastructures:** From dyson spheres to orbital rings, these titanic constructions dominate the space they occupy, often housing layers of cryptic mechanisms or hostile environments.
- **Energy Relics:** Floating, glowing nodes of immense power—often unstable—are found throughout their ruins, fueling speculation about their purpose.
- **Sentient Interfaces:** Some ruins house intelligences that may be remnants of Precursor minds or automated guardians. These entities often challenge visitors with puzzles, tests, or cryptic warnings.

Environmental Hazards

1. Dimensional Instability

Gravity wells, time loops, or spatial rifts plague many Precursor sites, making exploration perilous.

2. Automated Defenses

Drones, energy barriers, and traps designed to deter intruders may activate unexpectedly.

3. Corrupting Forces

Exposure to certain Precursor energy sources can alter organic and synthetic beings, sometimes granting power at a steep cost.

Precursor Artifacts

Types of Artifacts

- **Data Shards:** Encoded knowledge that requires advanced decryption to unlock. They might contain star maps, blueprints, or even warnings of cosmic events.
- **Resonance Devices:** Tools that manipulate energy, matter, or spacetime in ways modern technology cannot replicate. Examples include gravity projectors or devices that distort light and sound.
- **Living Relics:** Organic or semi-sentient artifacts that bond with their user, providing unique abilities but often influencing their thoughts and emotions.

Using Artifacts in Gameplay

1. Boosting Player Abilities

Artifacts can enhance characters' skills or grant unique powers, like temporary invisibility, energy shields, or advanced communication with alien entities.

2. Story Hooks

The discovery of an artifact can drive entire campaigns, whether it's a treasure hunt, a race against a rival faction, or an ethical dilemma over its proper use.

3. Complications

Many artifacts have unintended consequences. A resonance device might destabilize nearby systems, or a living relic might compel its user to act against their better judgment.

Mysteries of the Precursors

1. The Collapse

What caused their downfall? Every artifact and ruin is a piece of a puzzle that points toward an extinction event—one that might still pose a threat to the galaxy.

2. The Relic Guardians

Are the sentient constructs found in ruins merely ancient AI, or do they represent Precursor minds that refuse to die? Their cryptic warnings often mention a “Final Accord” — a phrase that appears in multiple unconnected ruins and has no confirmed translation. Some researchers believe it refers to the agreement the Precursors made before the end; others think it names the catastrophe itself. The Zynthari, when asked, do not answer.

3. The Shard Network

Some explorers believe that Precursor ruins are connected through an interstellar network of energy conduits. Activating one artifact could awaken others—or unleash chaos.

Random Artifact Table

Roll	1d10	for	inspiration:	d10	Arti-
fact	Description		—	-----	
-----				1	Shifting Lens A crystal that bends light to create illusions or invisibility. 2 Pulsar Key Unlocks hidden chambers in Precursor ruins but triggers defenses if misused. 3 Neural Imprint Sphere Implants fragments of Precursor knowledge into the user’s mind, at the risk of mental strain. 4 Temporal Stasis Rod Halts time within a small radius, useful for escape or sabotage. 5 Glyph Core Projects holographic maps or instructions, often incomplete or puzzling. 6 Void Anchor Stabilizes spacetime in unstable regions, allowing safe passage. 7 Luminal Seed Grows into a bio-mechanical construct over time, serving as an ally or resource. 8 Gravitic Prism Alters gravity in a small area, useful for traversal or combat. 9 Starforge Fragment Creates limited amounts of exotic materials when charged.

10 | Echo Beacon | Emits a signal that attracts both allies and enemies, and possibly awakens ancient constructs. |

Stellar Fauna and Flora

The alien worlds of the Orion Frontier support life forms adapted to conditions that range from toxic to impossible. Many are dangerous. Most are poorly understood. Below are examples of flora and fauna encountered in the Frontier.

Flora

1. Luminae Vines

- **Habitat:** Shadowed regions of dense Precursor ruins or deep cave systems.
- **Description:** These bioluminescent vines glow in mesmerizing hues, pulsating in response to movement or sound. Their tendrils are covered in tiny, reactive sensors, giving them the appearance of shimmering, living circuitry.
- **Behavior:**
 - The vines react defensively to perceived threats, releasing spores that induce hallucinations in intruders.
 - They coil around objects or living beings for protection, making exploration of their habitats treacherous.
- **Uses:** The glowing sap is sought after for its healing properties and as a source of light for frontier settlements, though harvesting it is highly dangerous.

2. Starlight Blooms

- **Habitat:** High-altitude plateaus on airless worlds.

- **Description:** These crystalline, flower-like structures are not plants in the traditional sense but silicon-based life forms that absorb starlight to fuel their growth. Each petal is semi-transparent and refracts light into dazzling rainbows.
- **Behavior:**
 - Blooms slowly rotate to track distant stars.
 - When disturbed, they emit a high-pitched frequency that can disrupt nearby electronics.
- **Uses:** Fragments of Starlight Blooms are prized as components in energy amplifiers and high-end optical technology.

Fauna

1. Umbra Stalkers

- **Habitat:** Subterranean caverns and shadowed planetary environments.
- **Description:** These sleek, six-legged predators are covered in a jet-black, reflective hide that renders them nearly invisible in darkness. They are bioluminescent on their undersides, emitting an eerie, low glow that they use to communicate with their kind.
- **Behavior:**
 - Ambush predators, they use their luminescence to lure prey into traps.
 - Highly intelligent, Umbra Stalkers have been observed coordinating hunts through complex light signals.
- **Threats:**
 - Capable of leaping great distances and attacking with razor-sharp mandibles.
 - Emit a disorienting chittering sound when threatened, which can incapacitate their prey.
- **Uses:** The reflective hide of an Umbra Stalker is a prized material for crafting stealth gear or armor. However, hunting them is highly dangerous, as their hunting packs are relentless in pursuit of perceived threats.

2. Nebula Mantas

- **Habitat:** Upper atmospheres of gas giants.
- **Description:** Enormous, manta-ray-like creatures that glide through gas giant skies, feeding on ionized particles and drifting organisms. Their translucent bodies shimmer with internal bioluminescence, creating the illusion of glowing constellations in motion.
- **Behavior:**
 - Passive filter feeders, but their size makes them dangerous to ships navigating their territory.
 - React to electrical discharges by emitting EMP pulses, disabling nearby electronics.
- **Threats:**
 - Accidental collisions with ships can cause catastrophic damage.
- **Uses:** Their unique bioluminescent enzymes are used in advanced sensor technology and stealth systems, making them highly valuable to researchers and corporations alike.

Ecosystems in the Frontier

Precursor Ruin Biomes

Precursor sites, with their energy anomalies and altered gravity fields, foster ecosystems unlike anything found elsewhere.

- **Symbiotic Life:** Many creatures and plants seem to have adapted to the residual energy of these sites, thriving where other organisms would perish.
- **Examples:**
 - **Glyph Crabs:** Small crustaceans with shell patterns resembling Precursor glyphs. They feed on the energy fields emitted by ruins.
 - **Echo Lichen:** A fungus that absorbs residual sound vibrations, turning them into energy. Colonies of Echo Lichen hum softly, creating an eerie ambient noise.

Gas Giant Skies

The dense, gaseous atmospheres of these worlds support entire airborne ecosystems.

- **Floating Ecosystems:** Creatures like the Nebula Mantas coexist with smaller filter-feeders and parasitic organisms.
- **Examples:**
 - **Sky Spires:** Towering, plant-like organisms that drift on air currents, releasing spores that act as seeds and food for other creatures.
 - **Ion Worms:** Tiny, glowing serpentine creatures that follow ion trails left by larger fauna, scavenging particles and organic debris.

Using Flora and Fauna in Gameplay

Adventure Hooks

1. Harvesting Danger

Players are hired to collect sap from Luminae Vines to treat an outbreak of radiation poisoning in a nearby colony. However, they must survive the hallucination-inducing spores and hostile wildlife.

2. Lost Expedition

A research team studying Starlight Blooms has gone missing on a barren world. The players must follow their trail and confront the creatures or environmental dangers responsible.

3. Umbra Stalker Infestation

A settlement's underground power plant has become the hunting ground for a pack of Umbra Stalkers. The players must eliminate the threat or find a way to drive the predators away.

4. Nebula Manta Harvesting

A corporation hires the players to extract bioluminescent enzymes from Nebula Mantas. Navigating the gas giant's treacherous skies while avoiding EMP pulses and corporate rivals proves to be no small task.

Random Encounter Table

Roll 1d10: | d10 | Encounter | | — | ----- | | 1 | A pack of Umbra Stalkers ambushes the players in a shadowed cave. | | 2 | A Luminae Vine snares an NPC, triggering a hallucinogenic response. | | 3 | A herd of Nebula Mantas passes dangerously close to the players’ ship, releasing EMP pulses. | | 4 | Glyph Crabs swarm the players’ equipment, attracted by its energy. | | 5 | A researcher studying Echo Lichen needs help deciphering its vibrations. | | 6 | Ion Worms infiltrate the players’ power systems, causing malfunctions. | | 7 | A rare bloom of Starlight Blossoms sparks a race between factions for their petals. | | 8 | A Sky Spire’s spores drift into a settlement, causing unknown biological effects. | | 9 | An alien nomad warns the players of a deadly creature haunting a Precursor ruin. | | 10 | A colossal Nebula Manta tangles with a pirate ship near the players’ position. |

Macro-Scale Hazards

Some threats transcend the scale of individual organisms and must be understood as system-level phenomena.

The Varnon Swarm

- **Scale:** Interstellar.
- **Classification:** Non-sentient self-replicating process; sometimes categorized as a distributed macro-organism, though the label is contested.
- **Description:** The Varnon Swarm does not have a body. It is a consuming process — a self-replicating cascade of structures that disassembles planetary matter, captures stellar energy, and replicates outward. Up close, an affected system shows signs of progressive mineral depletion, odd energy readings from orbit, and increasingly anomalous behavior from local fauna before planets become inert husks. The process takes centuries per system, but it does not stop.
- **Behavior:** No communication is possible. The Swarm does not pursue; it advances. Ships that remain in an actively consumed system too long report equipment failures consistent with energy siphoning. Direct contact with Swarm structures causes rapid material breakdown.

- **Uses:** The Swarm boundary is the setting for some of the most isolated and dangerous missions in the Orion Frontier. Researchers, early-warning teams, and reckless salvagers all venture there. First-hand observations are rare and valuable — most factions prefer their populations not to think too hard about what is moving toward them.

Mission Generator

This generator produces missions set in the Orion Arm. Players take on contracts for factions, uncover Precursor mysteries, and work through the competing interests of interstellar society.

Finding Missions

Roll 1d6 to determine the availability of missions. Spend 1¢ to re-roll.

D6	Job Offer
—	-----
1-2	No contracts available currently.
3-4	A risky or low-paying mission is offered.
5-6	Choose between two available contracts.

Mission Table

D20	Mission
—	-----
1.	Secure Precursor Artifact
2.	Investigate Alien Signal
3.	Establish Orbital Outpost
4.	Recover Lost Starship
5.	Assist Colony Infrastructure
6.	Neutralize Pirate Threat
7.	Protect Trade Convoy
8.	Survey Hazardous Planet
9.	Infiltrate Rival Facility
10.	Mediate Faction Dispute
11.	Capture Rogue AI Node
12.	Search for Missing Crew

- | 13. | Escort Diplomatic Envoy |
- | 14. | Test Advanced Weaponry |
- | 15. | Disable Smuggler Ring |
- | 16. | Aid in Terraforming Effort |
- | 17. | Repair Subspace Anomaly |
- | 18. | Contain Xenobiological Outbreak |
- | 19. | Reinforce Colony Defenses |
- | 20. | Discover New Resonance Point|

Location Table

- | D20 | Location |
- | — | ————— |
- | 1. | Core System Habitat |
- | 2. | Remote Frontier Colony |
- | 3. | Precursor Ruin Site |
- | 4. | Derelict Freighter |
- | 5. | Pirate Asteroid Base |
- | 6. | Gas Giant Mining Platform |
- | 7. | Volcanic Exoplanet |
- | 8. | Zynthari Border Zone |
- | 9. | Unstable Phase Rift |
- | 10. | Disputed Trade Hub |
- | 11. | Alien Biodome Station |
- | 12. | Neutron Star Proximity |
- | 13. | Uncharted Planetary System|
- | 14. | Cosmic Dust Nebula |
- | 15. | Rogue Planet Surface |
- | 16. | Varnon Swarm Boundary |
- | 17. | Experimental Shipyard |
- | 18. | Diplomatic Conference Station |
- | 19. | Deep Space Monitoring Array |
- | 20. | Isolated Terraforming Base|

Client Table

D20	Client
—	-----
1.	Sol Union High Command
2.	Orion Concordium Diplomatic Corps
3.	Sirius Compact Resource Authority
4.	Precursor Seekers Guild
5.	Independent Frontier Council
6.	Frontier Coalition Council
7.	Nova Terra Assembly
8.	Interstellar Free Market Consortium
9.	Zynthari Observation Directorate
10.	Rogue AI Research Division
11.	Colonial Infrastructure Union
12.	Artifact Protection Syndicate
13.	System Defense Coalition
14.	Pan-Galactic Traders Network
15.	Galactic Conservation League
16.	Zal-Terran Commonwealth Council
17.	Deep Space Xenology Institute
18.	Sector Energy Consortium
19.	Terraforming Engineers Assembly
20.	Independent Smuggler Broker

Complication Table

D20	Complication
—	-----
1.	Malfunctioning Starship Systems
2.	Rival Faction Interference
3.	Subspace Anomaly Disrupts Mission
4.	Hostile Xenobiological Entity
5.	Unstable Precursor Technology
6.	Crew Conflict or Betrayal
7.	Interstellar Communication Blackout

- | 8. | Sabotage by Unseen Forces |
- | 9. | Unexpected First Contact Event |
- | 10. | Xenological Contamination Risk |
- | 11. | Varnon Swarm Advance Detected Nearby |
- | 12. | Secret Corporate Involvement |
- | 13. | Factional Diplomatic Fallout |
- | 14. | Equipment Failure in Extreme Conditions |
- | 15. | Hostile Space Weather Phenomenon |
- | 16. | Misleading Mission Parameters |
- | 17. | Zynthari Intervention |
- | 18. | Rogue AI Opposition |
- | 19. | Unexpected Civilian Presence |
- | 20. | Distress Beacon is Hostile Trap |

Mission Reward Table

Missions pay according to their scope and danger. As a baseline: a **standard contract** pays 1d6¢ on completion (in addition to the Advancement d6¢ earned for personal growth); a **dangerous or faction-critical job** pays 2d6¢; an **exceptional outcome** or a job for a major polity yields 3d6¢ or an item from the table below instead of credits. The Referee should offer one or the other — credits or a table result — not both, unless the players have significantly exceeded mission parameters.

- | D20 | Reward |
- | — | ----- |
- | 1. | Substantial Financial Compensation |
- | 2. | Advanced Precursor Artifact |
- | 3. | Rare Trade Licenses |
- | 4. | Favor from Major Faction |
- | 5. | Unique Alien Biotechnologies |
- | 6. | Access to Restricted Star Systems |
- | 7. | Personal Recognition or Rank |
- | 8. | Experimental Starship Equipment |
- | 9. | Xenobiological Data Samples |
- | 10. | Outpost Ownership Rights |

- | 11. | Exclusive Research Access |
- | 12. | Valuable Star Map Coordinates |
- | 13. | Political Influence and Leverage |
- | 14. | Relic Deciphering Resources |
- | 15. | Enhanced Faction Standing |
- | 16. | Improved Starship Components |
- | 17. | Rare Geological Resources |
- | 18. | Unexplored Star System Data |
- | 19. | Precursor Knowledge Repository |
- | 20. | Alien Diplomatic Favor |

Twist Table

D20	Twist
1	The mission is a decoy for a larger plot
2	The client is revealed as untrustworthy
3	A crew member has secret connections to the mission
4	The target location is heavily booby-trapped
5	Unexpected alien involvement complicates the mission
6	A faction war erupts during the mission
7	A key ally betrays the team
8	The mission site becomes unstable
9	The artifact is sentient and resists capture
10	The client withdraws support midway
11	Another team is pursuing the same goal
12	Vital equipment malfunctions
13	A local rebellion interrupts progress
14	Time-sensitive conditions arise
15	The mission leads to an undisclosed treasure
16	The mission triggers a cosmic event
17	A rival crew intercepts the players

D20	Twist
18	The mission is revealed as a test by a major faction
19	Environmental hazards intensify
20	A hidden threat awakens

NPC Encounter Table

D20	Encounter
1	A desperate trader offering dubious goods
2	A mysterious Precursor scholar
3	A stranded pilot seeking assistance
4	An eccentric AI with unclear motives
5	A Zynthari diplomat carrying veiled warnings
6	A bounty hunter pursuing a shared target
7	A rogue faction leader offering an alliance
8	A corporate spy seeking intel
9	A rebellious settler seeking to escape
10	A seasoned smuggler with critical information
11	A zealot preaching Precursor divinity
12	A combat veteran seeking revenge
13	A rival scavenger with a cryptic clue
14	A pirate with a hidden agenda
15	A disgraced official seeking redemption
16	A hacker offering valuable tech
17	A shady merchant promising rare artifacts
18	A scientist obsessed with anomalies
19	A local leader pleading for aid
20	A covert operative gathering intel

Environmental Challenge Table

D20	Challenge
1	Intense radiation disrupts sensors
2	Gravity fluctuations impair mobility
3	Sudden electrical storms short-circuit equipment
4	Hostile native creatures attack
5	Toxic atmosphere requires constant monitoring
6	Extreme temperatures damage supplies
7	Collapsing terrain blocks progress
8	Unexpected gravitational pull shifts trajectories
9	Hidden traps within ruins
10	Explosive geothermal activity
11	Ice storms freeze critical systems
12	Dense asteroid field damages hull
13	Nebula interference blinds navigation
14	Rogue AI systems compromise security
15	Contaminated water supply spreads illness
16	Volcanic eruptions isolate the team
17	Dangerous wildlife stalks the crew
18	Unstable Precursor machinery activates
19	Solar flares disrupt communication
20	Alien spores compromise health

Faction Involvement Table

D20	Faction Influence
1	A secretive Zynthari envoy oversees the mission
2	Rival megacorporations vie for control of the outcome
3	The Precursor Seekers Guild offers cryptic guidance

D20	Faction Influence
4	A rogue AI network covertly assists, for unknown reasons
5	Frontier settlers resist interference from outsiders
6	A mercenary faction demands compensation for passage
7	The Sol Union dispatches operatives to monitor events
8	The Orion Concordium sends a diplomatic envoy
9	Smugglers exploit the mission site for profit
10	Religious zealots of the Fractured Path claim the mission site as sacred
11	Pirates use the mission as a distraction for larger raids
12	Nova Terra transhumanists intervene, seeing potential advancements
13	Rebel factions use the area as a staging ground for a larger attack
14	An alien species tries to reclaim its lost artifact or territory
15	The Eclipse Network manipulates events from behind the scenes
16	The Interstellar Free Market Consortium pressures for a favorable deal
17	Conservationists demand protection of local ecosystems
18	Rival archaeologists race to uncover a Precursor secret first
19	The Frontier Coalition demands a share of the spoils for cooperation
20	An unaligned collective of outlaws seeks leverage over all involved

Artifact Discovery Table

D20	Artifact Found
1	A hovering sphere containing encoded star charts
2	A gravity-defying obelisk with glowing glyphs
3	A shard of crystal that emits a melodic hum
4	A mechanical insect that appears semi-sentient

D20	Artifact Found
5	A disk that projects holographic schematics
6	A liquid-filled cylinder showing shifting forms
7	A lattice of unknown material absorbing nearby light
8	A floating cube that rearranges its faces continuously
9	A weapon-like device that seems too advanced for comprehension
10	A mask that connects to the wearer's mind, enhancing perception
11	A star-shaped pendant that radiates warmth in vacuum
12	A plant-like organism emitting faint electrical signals
13	A bracelet that generates a weak, impenetrable force field
14	A rod capable of disrupting electromagnetic fields nearby
15	A map showing Precursor ruins scattered across systems
16	A crystalline entity that interfaces with computer systems
17	A shard of metal that changes density at will
18	An organic-looking weapon that bonds with its user
19	A book-like artifact with shifting inscriptions
20	A sealed container that resists all attempts to open it

Precursor Ruins Features Table

D20	Ruin Feature
1	A chamber filled with levitating orbs
2	Walls that shift when touched
3	A holographic map of a distant galaxy
4	Pulsating glyphs that respond to movement
5	A massive spire emitting an unknown signal
6	A labyrinth that rearranges itself
7	A gravity-defying bridge
8	A vault containing suspended animation chambers

D20	Ruin Feature
9	A chasm filled with glowing mists
10	A control panel of incomprehensible complexity
11	Sentient guardians that challenge intruders
12	A reflecting pool that reveals alternate realities
13	A network of tunnels that seem endless
14	A room resonating with ancient songs
15	Crystal growths that shimmer in different frequencies
16	A suspended platform orbiting a mini black hole
17	An observatory showing celestial events millennia past
18	An artifact generator producing small anomalies
19	A floor etched with glowing alien constellations
20	A hidden sanctuary untouched by time

Space Travel Hazard Table

D20	Hazard Encountered
1	Unstable jumpgate disperses the ship's signal
2	Solar flares disable shields and electronics
3	A rogue asteroid cluster veers into the flight path
4	A gravity well from a collapsing star pulls the ship off course
5	A pirate ambush in a debris field
6	A temporal distortion warps the crew's perception of time
7	A ship-sized alien predator blocks the route
8	Subspace turbulence threatens to tear the ship apart
9	An AI virus spreads through navigation systems
10	A Zynthari border patrol stops the ship for questioning
11	A drifting wreck emits a distress signal
12	A void rift interferes with propulsion systems

D20	Hazard Encountered
13	A swarm of micrometeoroids damages hull integrity
14	An experimental weapon test zone proves unexpectedly active
15	Hostile weather in a gas giant's atmosphere disrupts sensors
16	An alien probe attaches itself to the hull
17	Space debris triggers ship-wide system malfunctions
18	A mysterious signal draws the crew off-course
19	Phase resonance misalignment causes a risky emergency jump
20	An interstellar convoy requests urgent aid against an ambush

NPC Role in Mission Table

D20	NPC Role
1	A reluctant informant hiding crucial intel
2	A saboteur working within the mission team
3	A mysterious benefactor funding the mission
4	A rival operative competing for the same objective
5	A pacifist with essential knowledge
6	A double agent playing both sides
7	A rogue trader offering crucial supplies
8	A skilled negotiator securing passage
9	A scientist obsessed with the mission's findings
10	A veteran pilot with invaluable experience
11	A reckless adventurer complicating plans
12	A covert assassin targeting a key player
13	A diplomat mediating between factions
14	An AI companion controlling vital systems
15	A Precursor fanatic jeopardizing the mission
16	A mercenary hired to protect the team

D20**NPC Role**

- | | |
|----|---|
| 17 | A scavenger holding a vital artifact |
| 18 | A fugitive seeking sanctuary during the mission |
| 19 | An injured settler with surprising expertise |
| 20 | A Zynthari observer silently judging every move |